# An Efficient Dual-Active-Bridge Converter for Wide Voltage Range Applications

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**Master of Technology**
**in Electrical Engineering with Specialization**
**in "Power and Energy Systems"**

*by*

**K.MOUNISH**

**DEPARTMENT OF ELECTRICAL ENGINEERING**
**INDIAN INSTITUTE OF TECHNOLOGY, KHARAGPUR**

**April 2026**

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Department of Electrical Engineering
Indian Institute of Technology Kharagpur
India - 721302

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## CERTIFICATE

This is to certify that I have examined the thesis entitled **"An Efficient Dual-Active-Bridge Converter for Wide Voltage Range Applications"**, submitted by **K.Mounish** (Roll Number: 24EE63R05), to Indian Institute of Technology, Kharagpur, India, is a record of bonafide Project work carried out by him under my supervision and guidance and is worthy of consideration for the award of the degree of Master of Technology in Electrical Engineering with Specialization in "power and energy systems" of the Institute.

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Dr. Deepak Reddy Pullaguram
Department of Electrical Engineering
Indian Institute of Technology Kharagpur

**Place:** Kharagpur
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**K. Mounish**
IIT Kharagpur

## ABSTRACT

The increasing penetration of renewable energy sources has introduced significant challenges due to their variability and intermittency, leading to energy curtailment during excess generation and power shortages at peak demand. On the other hand, retired electric vehicle (EV) batteries, though no longer suitable for vehicular use, retain considerable capacity and can be repurposed as second-life batteries (SLBs) in stationary energy storage systems (ESS), improving renewable energy utilization. Integrating such systems with the grid requires efficient Dual-directional power conversion over a wide voltage range.

The dual active bridge (DAB) converter is commonly used in applications that require bidirectional power transfer because it allows controlled energy flow between two DC sources also it has galvanic isolation, and soft-switching operation. However, conventional control techniques such as Single phase shift (SPS) suffer from high circulating current, limited zero-voltage switching (ZVS) range, and reduced efficiency under wide voltage variations. Advanced modulation methods including Extended phase shift (EPS), Dual phase shift (DPS), and Triple Phase Shift (TPS) offer improvements but remain insufficient for wide voltage range applications.

In this work, a comprehensive study of DAB converters is carried out, including modeling, simulation, and hardware validation. Bidirectional power flow is analyzed under closed-loop control using battery interfacing.The limitations of fixed transformer turns ratio under increased voltage and load conditions are investigated, showing increased RMS current, loss of ZVS, and efficiency degradation.

To address these challenges, a variable transformer equivalent turns ratio dual active bridge (VTDAB) converter is proposed. The proposed system utilizes dual transformers and modified switching strategies to achieve multiple effective turn ratios, allowing efficient operation across a wide voltage range. The design is validated through simulation, component selection, transformer design, and experimental verification using a multi-bridge hardware setup.

The results demonstrate improved performance in terms of reduced RMS current, extended ZVS range, and enhanced efficiency, making the proposed approach suitable for second-life battery-based energy storage applications.



**Keywords:** DAB, VTDAB, Wide Voltage Range, ZVS, RMS Current, Bidirectional Converter

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# Chapter 1

# Introduction

## 1.1 Background

Renewable energy sources such as solar and wind have expanded rapidly, reshaping modern power systems. However, their intermittent and non-dispatchable characteristics cause fluctuations in generation, resulting in an imbalance between supply and demand. Traditional power systems are designed to meet peak demand, even though such peaks occur only for short durations, resulting in inefficient utilization of infrastructure and increased operational costs.Maintaining real-time balance between generation and load has become increasingly challenging, especially with higher penetration of distributed generation .

Energy storage systems (ESS) play a vital role in addressing these challenges. ESS stores surplus energy during low-demand or high-generation periods and releases it during peak demand, enabling efficient peak shaving and load leveling. Energy storage also improves grid stability by delivering rapid response services such as frequency regulation, voltage support, and reserve capacity. It also reduces the stress on transmission and distribution networks and helps to delay costly infrastructure upgrades, which enhances the overall efficiency and reliability of the power system.

Despite these advantages, widespread deployment of battery energy storage systems is often limited by high costs, particularly for lithium-ion batteries. In this context, second-life electric vehicle (EV) batteries offer a promising and cost-effective alternative. Electric vehicle batteries are usually withdrawn from automotive use when their state of health decreases to around 70–80%. Although they still retain substantial usable capacity, these

batteries can be repurposed for stationary storage applications, providing services such as peak shaving, renewable energy buffering, and ancillary grid support. Not only reduces the cost of energy storage but also extends the life of the battery and minimizes environmental impact, making it a sustainable solution for future energy systems.

To effectively integrate such battery storage systems with the grid and renewable sources, bidirectional power converters play a crucial role. These converters enable controlled energy flow in both directions, allowing the battery to charge during excess generation and discharge during high demand. Advanced converter topologies, Converters like the Dual Active Bridge (DAB) are well suited for such applications because they offer high efficiency, provide galvanic isolation, and allow precise control of power flow through phase-shift modulation. In addition, bidirectional converters support essential grid functions, such as voltage regulation, frequency stabilization, and improved power quality. They also enable advanced applications such as vehicle-to-grid (V2G) systems and distributed energy integration. Figure 1.1 illustrates the configuration of a microgrid system.

Therefore, the combined use of energy storage, second-life EV batteries, and bidirectional converters forms a comprehensive solution to address the challenges posed by renewable energy integration.This approach improves grid flexibility. It also reduces cost and supports sustainable energy integration.

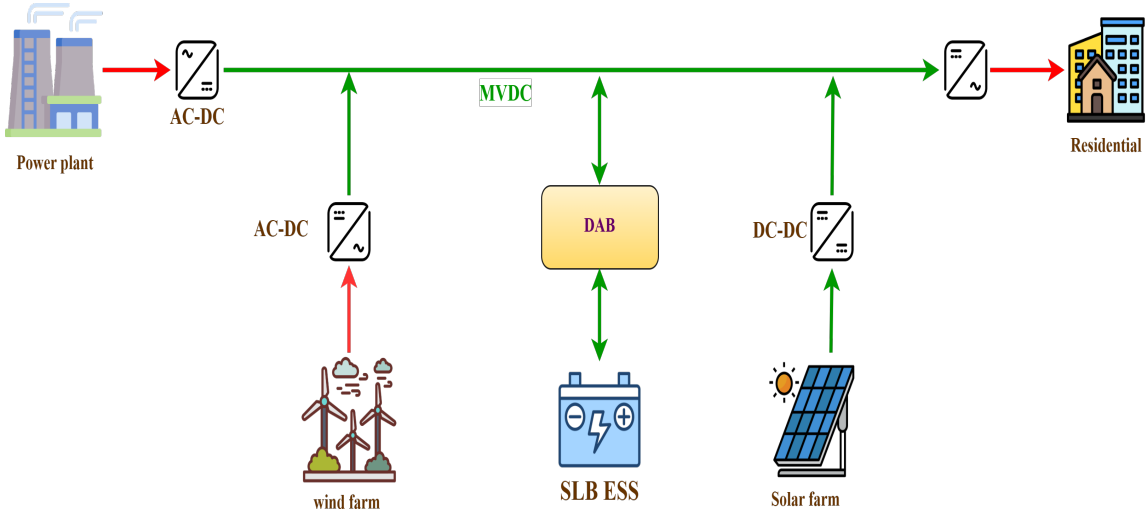



Figure 1.1: DC Microgrid with Renewable Energy Sources and SLB ESS

## 1.2 Problem Statement

The increasing penetration of renewable energy sources and energy storage in modern grids requires bidirectional DC-DC converters that can operate efficiently across a wide voltage range. The dual active bridge (DAB) converter has become a suitable topology for such applications owing to its galvanic isolation, inherent bidirectional power flow capability, and compatibility with soft-switching techniques. However, despite these advantages, several fundamental challenges limit the practical deployment in conventional DAB (CDAB) converters when operating over a wide voltage range.

The effectiveness of conventional SPS control largely depends on voltage matching conditions. When the input and output voltages deviate, efficiency drops due to increased RMS current, where the product of the transformer turn ratio and the secondary voltage matches the primary voltage ($nV_2 = V_1$). When the operating voltage deviates significantly from this matched point which is inherently the case with second-life EV batteries whose terminal voltage varies across their state of charge. Further, the inductor RMS current rises sharply, exceeding twice its matched-point value at lower voltages. This results in substantially increased conduction losses and copper losses, directly degrading converter efficiency.

ZVS is critical for minimizing switching losses in high frequency DAB converters. The CDAB achieves full ZVS for all switches only in the vicinity of the voltage matched condition. As the secondary voltage deviates either below or above the matching point specific switches lose ZVS, leading to hard switching and significant switching losses.

Advanced modulation methods like extended phase shift (EPS), dual phase shift (DPS), and triple phase shift (TPS) minimize circulating current, reactive power, and current stress compared to SPS by incorporating additional internal phase shifts within the full bridges.However, these improvements are effective only within a moderate voltage variation range. When the voltage gain range is widened significantly (e.g., 0.4 to 2.5 as required for second-life battery ESS), the benefits of phase shift control diminish considerably and adjustments to the DAB topology itself become necessary.

The fundamental limitation of the CDAB in wide voltage range operation is its fixed transformer turns ratio. With a fixed turns ratio, the converter cannot adapt its voltage matching point to suit different operating voltages, and no modulation strategy alone can compensate for the large voltage mismatch across the full operating range. Existing solutions such as relay based bridge reconfiguration, auxiliary half-bridge circuits introduce

additional components, increase circuit complexity, increase cost, and complicate control implementation.

In summary, there exists a clear need for a DAB converter topology that can achieve multiple effective transformer turns ratios across a wide voltage gain range without auxiliary components such as relays or extra switches while maintaining low inductor RMS current, wide ZVS range, and high efficiency for both charging and discharging operation. This challenge is addressed by the proposed variable transformer equivalent turns ratio dual active bridge (VTDAB) converter presented in this work.

## 1.3 Objectives

The key objectives of this work are outlined as follows:

- To examine the limitations of the CDAB converter under wide voltage operating conditions,with a focus on the increase in inductor RMS current, reactive power, switching losses, and the loss of ZVS at non-nominal voltage ratios.
- To investigate and comparatively evaluate advanced phase-shift modulation strategies, namely EPS, DPS, and TPS, in terms of their ability to minimize circulating current and expand the ZVS operating range,and improve overall conversion efficiency.
- To develop a variable transformer equivalent turns ratio dual active bridge (VTDAB) topology employing dual transformers and optimal phase shift (OPS) control for efficient wide-voltage-range operation suitable for second-life electric vehicle (EV) battery applications.
- To validate the proposed VT-DAB system through detailed simulation studies, including closed-loop bidirectional power control for both grid-to-battery and battery-to-grid operation,and to compare its performance with conventional and advanced DAB control techniques.
- To experimentally validate the proposed VTDAB topology using two Voltage Source Inverters (VSI) available in the laboratory, with two legs of one VSI and three legs of the other VSI forming the required VTDAB converter bridges, integrated with the custom-designed dual-transformer setup, and to evaluate performance across the wide voltage operating range.

# Chapter 2

# Literature Study

## 2.1 DC–DC Converters

DC-DC converters form the backbone in modern power electronic systems.They are used to convert voltage levels while maintaining control over power flow. This conversion is achieved through the controlled switching of semiconductor devices, where energy is periodically stored in passive elements such as inductors, capacitors, or transformers and then delivered to the output at the required voltage level.

In practical applications, these converters are not limited to simple voltage scaling. They also play an important role in controlling power flow, improving power quality, and enabling compatibility between different subsystems operating at different voltage levels. This makes them indispensable in systems involving renewable energy sources, energy storage units, and DC distribution networks.

Depending on the direction of power transfer, DC–DC converters can operate either in unidirectional or bidirectional modes. Unidirectional converters allow power flow in only one direction and are typically used in applications such as photovoltaic generation, where energy is delivered from the source to the load. In contrast, bidirectional converters enable energy exchange in both directions, making them essential for systems like battery storage, where charging and discharging operations are required.

Bidirectional DC–DC converters have become increasingly important in recent years due to their broad range of applications. They are commonly used in renewable energy systems, electric and hybrid vehicles, aerospace systems, and DC microgrids. In these applications,

they serve as an interface between energy sources and storage elements, allowing efficient energy transfer and dynamic control of the power flow.

From a structural from a practical perspective, bidirectional converters can be generally categorized into non-isolated (NIBDC) and isolated (IBDC) topologies, as shown in Fig. 2.1. NIBDC converters, including buck-boost and interleaved types, offer a simple structure but lack electrical isolation and are unsuitable for high-power applications. IBDC converters overcome this using high-frequency transformers for galvanic isolation and high efficiency voltage conversion.

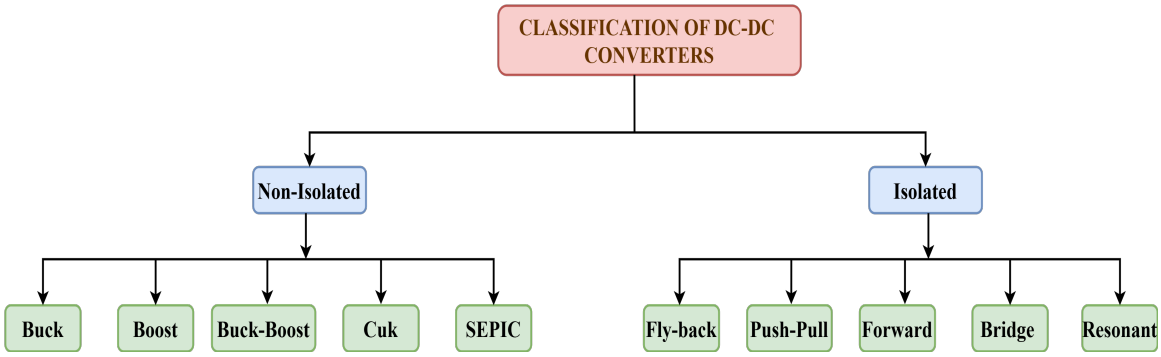



Figure 2.1: Classification of DC-DC converters

In contrast, isolated converters use a high-frequency transformer between the input and output stages. This allows galvanic isolation, improves safety, and enables flexible voltage conversion through transformer turns ratio. Although this adds complexity and increases design effort, isolated topologies are commonly preferred in applications where high power levels, safety, and reliability are critical.

Another important aspect of DC–DC converters is their ability to control power transfer through modulation techniques. By adjusting switching patterns such as the output voltage can be controlled by adjusting the duty cycle or phase shift of the converter, control current, and manage the direction of power flow. Advanced modulation methods are especially important in bidirectional converters, where efficient operation must be maintained under varying load and voltage conditions.

Overall, DC–DC converters form the backbone of modern DC-based energy systems. Their versatility, efficiency, and controllability make them essential for the integration of renewable energy sources, managing energy storage, and enabling the development of advanced power systems such as DC microgrids.

## 2.2 Dual Active Bridge Converter

The Dual Active Bridge converter is an auto-adjustable, bidirectional, isolated DC-DC converter. It draws attention because of its specific features such as galvanic isolation, zvs, wide voltage gain range, and high efficiency. Even though there are several alternatives among IBDC topologies such as Flyback, Cuk, SEPIC, and LLC resonant converters these features make it singular.

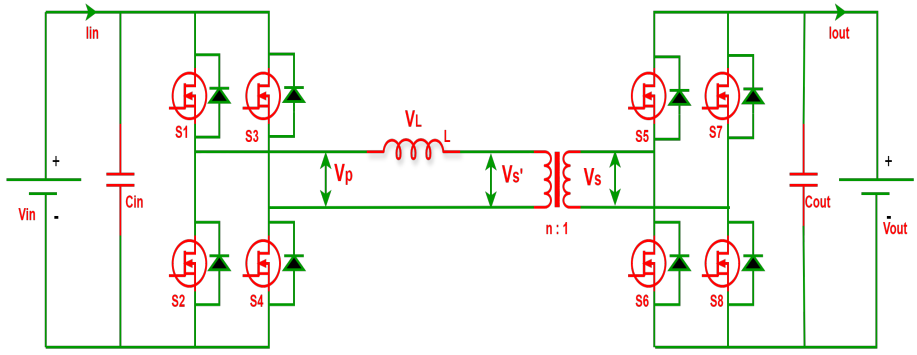



Figure 2.2: Dual Active Bridge Converter

The DAB consists of two H-bridges separated by an inductor and a high-frequency transformer. The topology of the DAB is shown in Figure 2.2. Power transfer in the DAB is achieved by introducing a phase shift between the PWM signals of both bridges. By tuning this phase shift DAB controls the direction and magnitude of power flow from the leading bridge to the lagging bridge, which is reversible simply by reversing the sign of the phase shift. Input and output filter capacitors $C_{in}$ and $C_{out}$ are connected at the respective DC buses to suppress the voltage ripple.

## 2.3 Steady State Operation

For steady-state analysis, the DAB is represented by the simplified equivalent circuit as shown in Figure 2.3 where all quantities are referred to the primary side, transformer parasitics other than leakage inductance are neglected.Here in equivalent DAB where $V_{ac1} = n\,V_{ac2}$ and $L$ represent the total circuit inductance.

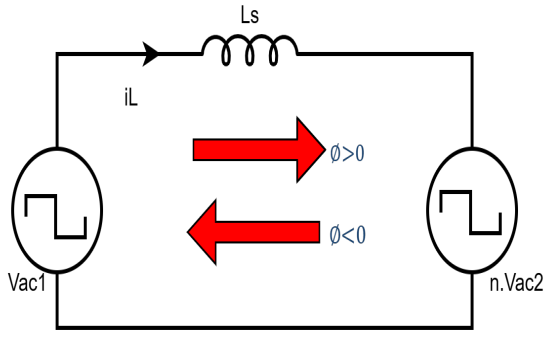



Figure 2.3: Equivalent circuit of DAB

The instantaneous inductor current at any time instant $t_1$ is determined by integrating the net voltage across the inductance over the preceding interval.

$$i_L(t_1) = i_L(t_0) + \frac{1}{L_s}\int_{t_0}^{t_1} \left[v_{ac,1}(t) - n\, v_{ac,2}(t)\right] dt \tag{2.1}$$

Equation (2.1) can be reformulated in its differential form, as given in Equation (2.2). This representation reveals that the rate of change in inductor current at any instant is proportional to the difference between the two referred bridge voltages and inversely proportional to the inductance $L_s$.

$$\frac{di_L(t)}{dt} = \frac{v_{ac,1}(t) - n\, v_{ac,2}(t)}{L_s} \tag{2.2}$$

The average power delivered by the converter can be evaluated from the inductor current $i_L$, as formulated in Equations (2.3) and (2.4), where $T_s$ is the switching period corresponding to a switching frequency $f_s = \frac{1}{T_s}$.

$$P_1 = \frac{1}{T_{s1}}\int_0^{T_{s1}} P_1(t)\, dt = \frac{1}{T_{s1}}\int_0^{T_{s1}} v_{ac,1}(t)\, i_L(t)\, dt \tag{2.3}$$

$$P_2 = \frac{1}{T_{s2}}\int_0^{T_{s2}} P_2(t)\, dt = \frac{1}{T_{s2}}\int_0^{T_{s2}} v_{ac,2}(t)\, i_L(t)\, dt \tag{2.4}$$

Under lossless operating conditions, conservation of energy requires the input and output power to be equal, as stated in Equation (2.5a). It follows from this condition, as shown in Equation (2.5b), that the secondary current $i_2(t)$ is a turns ratio scaled form of the primary

current $i_1(t)$. Consequently, within a lossless framework, $i_L$ also serves as a representation of the referred secondary current.

$$P = |P_1| = |P_2| \tag{2.5a}$$

$$n\, i_1(t) = i_2(t) \tag{2.5b}$$

The amount of power transferred through the DAB converter is dictated by the properties of the square wave voltages applied to both bridges, including their amplitudes, duty cycles, switching frequency, and the phase relationship between them.

## 2.4 Phase Shift Modulation

The phase shift represents the time delay between the switching instants of the primary and secondary bridges. By varying this phase shift, the DAB converter regulates both the direction and magnitude of power flow. Figure 2.4 illustrates the phase shifts $D0$, $D1$, and $D2$ applied to the primary and secondary bridges of the DAB converter.

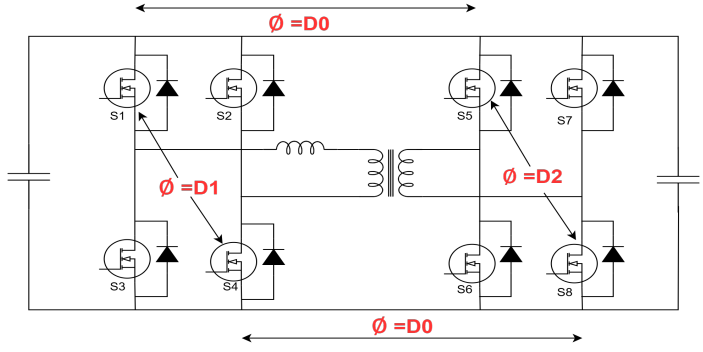



Figure 2.4: Switching states corresponding to phase shifts ($D_0$, $D_1$, and $D_2$)

There are four major phase shift modulation techniques used in DAB:

| | | |
|---|---|---|
| Single phase shift (SPS) | $\Rightarrow$ | $D_1 = D_2 = 0,\ D_0 \neq 0$ |
| Extended phase shift (EPS) | $\Rightarrow$ | $D_1 \neq 0,\ D_2 = 0$ |
| Dual phase shift (DPS) | $\Rightarrow$ | $D_2 \neq 0,\ D_1 = D_2,\ D_0 \neq 0$ |
| Triple phase shift (TPS) | $\Rightarrow$ | $D_1 \neq 0,\ D_2 \neq 0,\ D_0 \neq 0$ |
| Optimal phase shift (OPS) | $\Rightarrow$ | Combination of SPS, EPS, DPS, and TPS |

# Chapter 3

# Working of DAB with Different Phase Shift Techniques

## 3.1 Working of DAB under SPS

SPS is a simple and widely used phase shift control method. It has a single degree of freedom, namely the external phase shift

$$(D_0)$$

, which determines the power transfer. The primary side is supplied by a DC voltage source that delivers power to the secondary side. Based on the application, the secondary side load may be either resistive or represented as a voltage source.

The presence of the inductor and switch capacitance's and the resonation between them make it possible to achieve zero voltage switching which is clearly explained in further stages. Higher inductance is needed for low switching frequencies. Power transfer relies on leakage inductance and phase shift angle.

In SPS DAB, he primary and secondary bridges are driven in synchronism, with all switches operating at a 50% duty cycle. The diagonal switch pairs are switched together, producing a square-wave output from each bridge. To derive the voltage transmission equations for the primary and secondary sides, the inductor voltage and current are analyzed.The inductor voltage $V_L$ is given by the difference between voltage at the primary side ($V_p$) and the voltage converted to the primary side ($V_s'$). Here, $V_s'$ represents the secondary voltage

referred to the primary side through the transformer turns ratio $n$. Similarly, $V_o'$ denotes the output voltage $V_o$ reflected to the primary side.

Because of the phase difference between the primary and secondary sides sides, the inductor voltage waveform exhibits three distinct levels, leading to four operating intervals under different switching conditions. The corresponding inductor voltage values are given by $(V_{in} + nV_o)$, $(V_{in} - nV_o)$, $(-V_{in} - nV_o)$, and $(-V_{in} + nV_o)$ which are obtained from the interval 1,2,3,4.

### 3.1.1 Switching Sequence of DAB in SPS

The switching process is split into four intervals according to the inductor current behavior and the phase difference between the primary and secondary sides of the transformer voltages explained in figure 3.1.

**Interval 1** $(0 \leq t < T_\phi)$**:** Switches $S_1S_4$ are on the primary side and apply $+V_1$, while $S_6S_7$ are on the secondary and apply $-V_2$. The inductor experiences a net voltage of $(V_1 + V_2)$, where $V_1 = V_{in}$ and $V_2 = -nV_o$, resulting in:

$$L\frac{di_L(t)}{dt} = V_1 + V_2 = V_{in} + nV_o \tag{3.1}$$

Due to the phase-shift transition (dead-time), the secondary bridge commutates to $S_5S_8$, making the secondary voltage $+V_2$. During this interval, the inductor current crosses zero and contains both positive and negative components.

**Interval 2** $(T_\phi \leq t < T_s/2)$**:** Primary remains at $+V_1$ $(S_1S_4)$ and secondary is $+V_2$ $(S_5S_8)$; the inductor voltage reduces to $(V_1 - V_2)$, where $V_1 = V_{in}$ and $V_2 = nV_o$, hence:

$$L\frac{di_L(t)}{dt} = V_1 - V_2 = V_{in} - nV_o \tag{3.2}$$

**Interval 3** $(T_s/2 \leq t < T_s/2 + T_\phi)$**:** Primary switches to $-V_1$ $(S_2S_3)$ while secondary initially remains at $+V_2$ $(S_5S_8)$ and later transitions to $-V_2$ $(S_6S_7)$; the inductor sees $(-V_1 - V_2)$, where $V_1 = -V_{in}$ and $V_2 = nV_o$, giving:

$$L\frac{di_L(t)}{dt} = -V_1 - V_2 = -V_{in} - nV_o \tag{3.3}$$

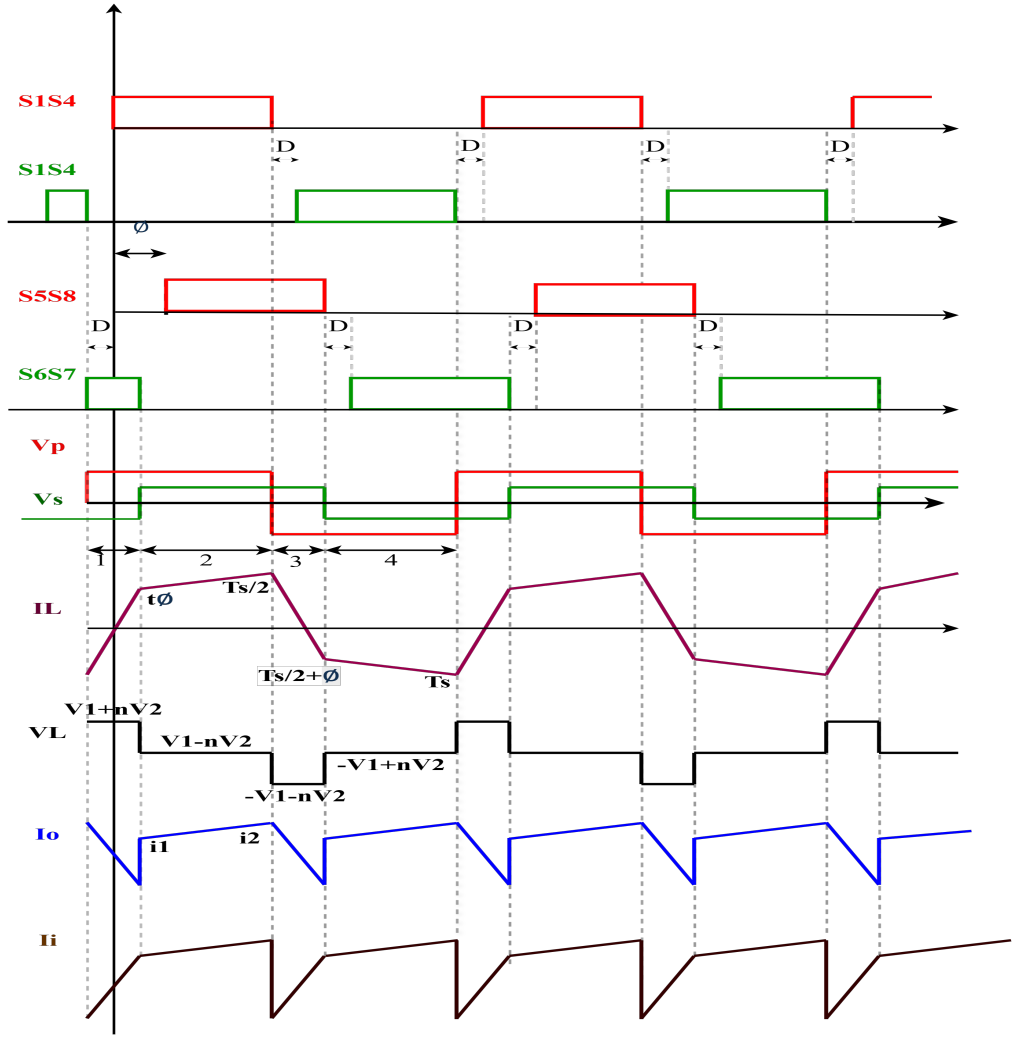



Figure 3.1: Switching diagram for DAB converter,SPS

Again, due to commutation, the current includes both positive and negative values.

**Interval 4** ($T_s/2 + T_\phi \leq t < T_s$)**:** Primary remains at $-V_1$ ($S_2S_3$) and secondary is $-V_2$ ($S_6S_7$). The inductor voltage becomes $(-V_1 + V_2)$, where $V_1 = -V_{in}$ and $V_2 = -nV_o$, hence:

$$L\frac{di_L(t)}{dt} = -V_1 + V_2 = -V_{in} + nV_o \tag{3.4}$$

**From Time Interval I:**

$$i_L(t) = i_L(0) + \frac{(V_{dc} + nV_o)t}{L} \quad ; \; 0 < t < T_\phi \tag{3.5}$$

**From Time Interval II:**

$$i_L(t) = i_L(T_\phi) + \frac{(V_{dc} - nV_o)(t - T_\phi)}{L} \quad ; \; T_\phi < t < \frac{T_S}{2} \tag{3.6}$$

Due to the half-cycle symmetry and with $T_\phi = \frac{\phi}{2\pi f_S}$

$$i_L(0) = \frac{\pi(nV_o - V_{dc}) - 2\phi nV_o}{4\pi f_S L} \tag{3.7}$$

which results for a positive phase shift, $0 < \phi < \pi$. A similar expression is obtained for negative phase shift, $-\pi < \phi < 0$.

In steady state, the output power $P_o$ is determined by the product of the output voltage and the average rectifier current, i.e.,

$$P_o = V_o \bar{i}_r(t)$$

The waveform of $i_o$ used for this evaluation is shown in Figure 3.1.

$$P_o = \frac{n\,V_{dc}\,V_o\,\phi\,(\pi - \phi)}{2\pi^2 f_S L} \qquad ; \; 0 < \phi < \pi \tag{3.8}$$

Similarly

$$P_o = \frac{n\,V_{dc}\,V_o\,\phi\;(\pi - |\phi|)}{2\pi^2 f_S L} \qquad ; \; -\pi < \phi < \pi \tag{3.9}$$

$$\phi = \frac{\pi}{2}\left[1 - \sqrt{1 - \frac{8 f_S L |P_o|}{nV_{dc}V_o}}\right] \mathrm{sgn}(P_o) \qquad ; \; |P_o| < |P_{PS,\max}| \tag{3.10}$$

### 3.1.2 Zero Voltage Switching

Switching loss arises during the transition of the switching device between its on and off states, coinciding with the intersection of the switch voltage and current flow. To mitigate dynamic dissipation, soft switching techniques are employed. Ideally, soft switching entails the device transitioning between on and off states at zero current or zero voltage. In practice, these transitions occur at low voltage or current levels, resulting in minimal dissipation. Soft switching converters are broadly categorized into three main types: resonant converters,

resonant switch converters, and phase-shifted converters.

Resonant converters necessitate additional components and adjustment of the switching frequency to regulate power transfer and achieve soft switching. In contrast, phase-shifted converters capitalize on the inherent parasitic elements to facilitate soft switching, while adjusting the phase shift to modulate power transfer. Consequently, in phase-shifted bridge converters, the control system harnesses the parasitic elements of the primary components transistors and transformer to facilitate smooth transitions. Similar to conventional converters, phase-shifted converters operate at a consistent frequency with reduced voltage and current stresses. Thus, during transition modes, phase-shifted converters exhibit resonant converter-like behavior, while functioning related to conventional converters in other modes.

The DAB converter is renowned for its intrinsic soft-switching capability, requiring no additional auxiliary components. Achieving Zero Voltage Switching (ZVS) leverages the leakage inductance of the transformer and the parasitic capacitance of the MOSFETs, along with their parallel body diodes. To ensure ZVS in both the leading and lagging full bridges, it is imperative that the currents through the leakage inductances at the switching instant, $i_1$ and $i_2$ respectively, be greater than zero. This condition enables current flow through the parasitic body diodes of the transistors to facilitate a soft-switching turn-on.

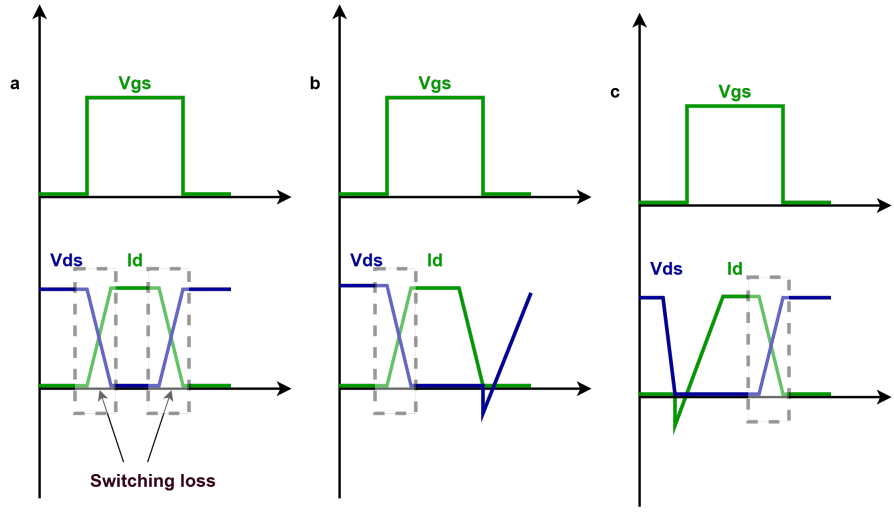



Figure 3.2: a) Hard switching b) Zero current switching c) Zero voltage switching

**Dead time:** Dead time is the intentional "blanking" interval inserted between turning **off** one device and turning **on** its complementary device, to prevent shoot through.

- During the dead-time, the energy stored in the leakage inductance free-wheels through

the body diodes and resonates with the MOSFETs' parasitic output capacitances.

- If dead time is 0, it may still be possible to get ZCS but never ZVS. During $t_{off}$, current naturally goes to 0.

**ZVS transition in secondary side from interval 1 to 2:**

- When the operation shifts from the first interval to the second, the primary-side switches

  $Q_1$ and $Q_4$ remain in conduction, while on the secondary side,$Q_6$ and $Q_7$ turn off and $Q_5$ and $Q_8$ turn on.

- The voltage across $Q_6$ and $Q_7$ is zero during conduction, while $Q_5$ and $Q_8$ withstand the full secondary voltage.

- During the dead time, when all secondary-side switches are off, the energy stored in the inductor sustains the current, discharging the capacitors across MOSFETs $Q_5$ and $Q_8$ to zero, while simultaneously charging the capacitors of $Q_6$ and $Q_7$ to the full secondary voltage.

- After the capacitors are fully charged and discharged, the current continues to flow through diodes $D_5$ and $D_8$, which clamp the voltage across MOSFETs $Q_5$ and $Q_8$ to zero. In the following interval,$Q_5$ and $Q_8$ urn on under zero-voltage conditions, eliminating turn-on losses.

- The inductor's stored energy discharges the MOSFET output capacitances, keeping their voltage near zero before turn-on. This condition, where the device voltage is nearly zero during switching, is known as zero-voltage switching(**ZVS**).

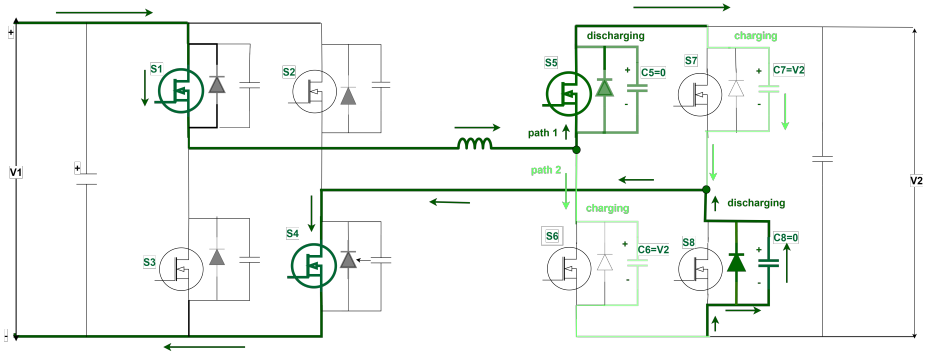



Figure 3.3: ZVS transition — secondary side interval 1 to 2

**ZVS transition in primary side from interval 2 to 3:**

- When the operation shifts from the second to the third interval, the secondary-side switches $Q_5$ and $Q_8$ remain conducting, while on the primary side $Q_1$ and $Q_4$ turn off and $Q_2$ and $Q_3$ turn on.
- Initially, the voltage across $Q_1$ and $Q_4$ is zero during conduction, while $Q_2$ and $Q_3$ withstand the full primary voltage.
- During the dead time, when all primary-side switches are off, the energy stored in the inductor maintains the current, discharging the capacitors of MOSFETs $Q_2$ and $Q_3$ to zero while charging those of $Q_1$ and $Q_4$ to the full primary voltage.
- After the capacitors are fully charged and discharged, the current continues through diodes $D_2$ and $D_3$, which clamp the voltage across MOSFETs $Q_2$ and $Q_3$ to zero.
- In the following interval, MOSFETs $Q_2$ and $Q_3$ switch on at zero voltage, eliminating turn-on losses.

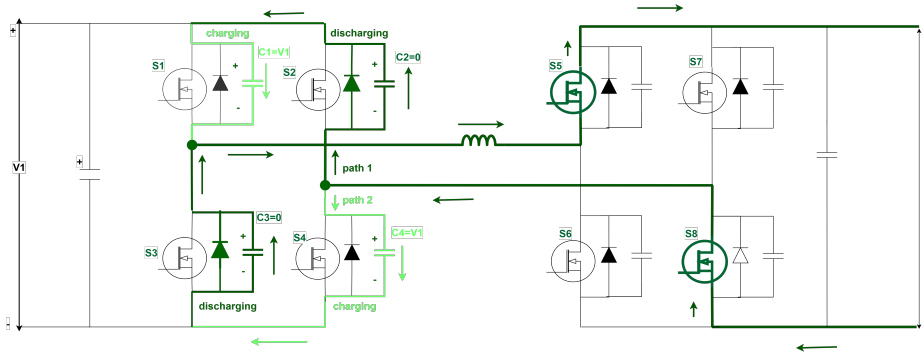



Figure 3.4: ZVS transition — primary side interval 2 to 3

### 3.1.3 Conditions for zero voltage switching

The converter's voltage gain $M$ defines the relationship between the output voltage referred to the primary side and the input voltage, where $n$ is the transformer turns ratio $\left(n = \frac{N_2}{N_1}\right)$:

$$M = \frac{nV_2}{V_1} \tag{3.11}$$

- For ZVS during the transition from interval one to interval two,$I_L$ must be positive when $Q_5$ and $Q_8$ are conducting; this current is denoted as $i_1$. Similarly, for the transition from interval two to three, $I_L$ust also remain positive, referred to as $i_2$.
- For ZVS transition between interval three and interval four, and interval four and interval one, $I_L$ needs to be negative.
- Dead time should be greater than the resonant time.
- Resonant frequency must be greater than switching frequency.
- Minimum inductor current to achieve ZVS is:

$$I_{L,min} = \sqrt{\frac{C_T \cdot V_T^2}{L}} \tag{3.12}$$

  where $C_T$ is the transistor output capacitance and $V_T$ is the transistor voltage, both available in datasheets.
- When $M = 1$, $i_1$ and $i_2$ are equal and positive for positive phase shift (light loads).
- When $M \neq 1$ and $M < 1$, then the lagging bridge experiences hard switching. ZVS is achieved only if:

$$D_\phi > \frac{1 - M}{2} \tag{3.13}$$

- When $M \neq 1$ and $M > 1$, then the leading bridge experiences hard switching. ZVS is achieved only if:

$$D_\phi > \frac{M - 1}{2M} \tag{3.14}$$

At higher phase-shift values, corresponding to higher current and power, ZVS is attained even for $M$ values significantly different from 1. Conversely, for low $D_\phi$ values, ZVS is only realised when $M$ is approximately 1.

### 3.1.4 Limitations of DAB under SPS

The conventional Single Phase Shift (SPS) control method suffers from several limitations, particularly under wide voltage range operation:

- Large backflow power (BFP) when the voltage conversion ratio $k = \frac{V_1}{nV_2}$ deviates significantly from unity.

- High current stress and increased inductor RMS current.
- Narrow Zero Voltage Switching (ZVS) range.
- Large circulating power, leading to increased conduction and switching losses.
- Poor efficiency under light load conditions and wide voltage variations.
- Surge current peaks reaching 3–5 times the steady-state value during startup.

To overcome these limitations, Extended Phase Shift (EPS) control was proposed.

## 3.2 Working of DAB under EPS

Extended Phase Shift (EPS) control introduces an additional degree of freedom compared to SPS by applying an inner phase shift $D_1$ within the primary bridge, resulting in two degrees of freedom overall.

Unlike SPS, the primary voltage $V_1$ is a 2-level square wave ($\pm V_1$). In EPS, the primary voltage $v_1$ becomes a 3-level quasi-square wave ($V_1,\ 0,\ -V_1$) due to $D_1$, and the secondary voltage $v_2$ remains a 2-level square wave with 50% duty ratio. This additional inner phase shift allows the timing between the two legs within one bridge to be controlled independently before shifting the entire secondary bridge relative to the primary, enabling the current waveform in the leakage inductor to be precisely sculpted.

This 3-level primary voltage fundamentally changes the inductor current waveform and reduces the duration and magnitude of backflow power. As a result of this additional phase shift, the inductor voltage in EPS exhibits six distinct levels across one switching cycle, namely $nV_o,\ (V_{in} + nV_o),\ (V_{in} - nV_o),\ -nV_o,\ (-V_{in} - nV_o),\ (-V_{in} + nV_o)$

The primary benefit of EPS over SPS is the reduction of RMS current in the leakage inductor, which directly lowers conduction losses and improves overall converter efficiency, particularly under voltage mismatch conditions where SPS performance degrades significantly.

### 3.2.1 Switching Sequence of DAB in EPS

The converter operation under EPS can be divided into six modes as shown in Figure 3.5.

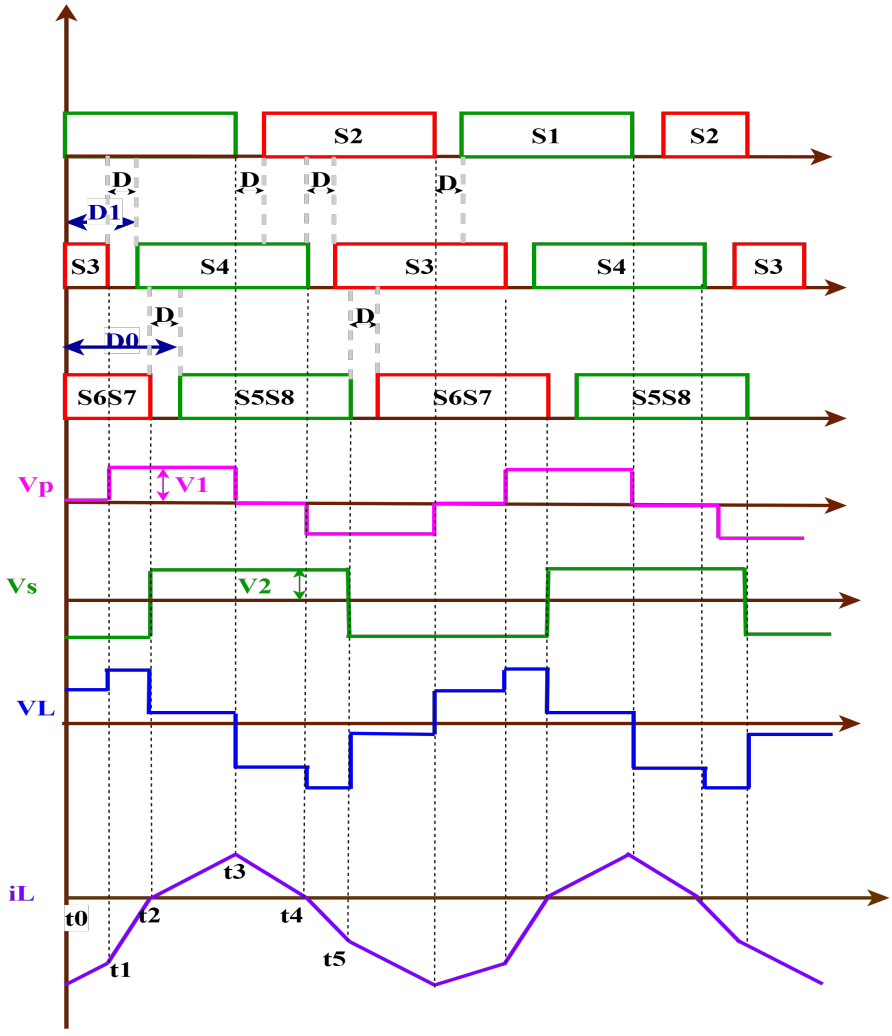



Figure 3.5: Switching diagram for DAB converter, EPS

**mode 1** $(0 \leq t < T_\phi)$**:** Before $t_0$, $S_2S_3$ are conducted and the current flows in the negative direction. At time $t_0$, $S_1$ turns on but unlike SPS, $S_4$ is not on with $S_1$ but is shifted, so the freewheeling path is provided by $S_3$ and $D_1$, resulting in $V_{in} = 0$ on the primary side. On the secondary side $S_6S_7$ are conducted giving $-V_o$, where $V_1 = 0$ and $V_2 = -nV_o$, resulting in:

$$L\frac{di_L(t)}{dt} = V_1 + V_2 = nV_o$$

**mode 2** $(T_\phi \leq t < T_\phi + T_{\phi_o})$**:** At time $t_1$, the current is still negative. Switch $S_3$ turns off and $D_4$ turns on, resulting in $+V_{in}$ on the primary side. The secondary side $S_6S_7$ remains

on giving $-V_o$, where $V_1 = V_{in}$ and $V_2 = -nV_o$, hence:

$$L\frac{di_L(t)}{dt} = V_1 + V_2 = V_{in} + nV_o$$

**mode 3** $(T_\phi + T_{\phi_o} \leq t < T_s/2)$**:** At time $t_2$, $S_6S_7$ turn off and $D_5D_8$ turn on giving $+V_o$ on the secondary side. On the primary side $S_1S_4$ are on, where $V_1 = V_{in}$ and $V_2 = nV_o$, giving:

$$L\frac{di_L(t)}{dt} = V_1 - V_2 = V_{in} - nV_o$$

**mode 4** $(T_s/2 \leq t < T_s/2 + T_\phi)$**:** At time $t_3$, $S_1$ turns off and $D_2$ turns on while $S_4$ remains on, so freewheeling occurs from $S_4$ to $D_2$. Since the current direction is positive, $V_{in} = 0$ on the primary side. On the secondary side $S_5S_8$ turn on giving $+V_o$, where $V_1 = 0$ and $V_2 = nV_o$, hence:

$$L\frac{di_L(t)}{dt} = -V_2 = -nV_o$$

**mode 5** $(T_s/2 + T_\phi \leq t < T_s/2 + T_\phi + T_{\phi_o})$**:** At time $t_4$, $S_4$ turns off and $D_3$ turns on while $S_2$ remains on, resulting in $-V_{in}$ on the primary side. The secondary side $S_5S_8$ remain on giving $+V_o$, where $V_1 = -V_{in}$ and $V_2 = nV_o$, giving:

$$L\frac{di_L(t)}{dt} = -V_1 - V_2 = -V_{in} - nV_o$$

**mode 6** $(T_s/2 + T_\phi + T_{\phi_o} \leq t < T_s)$**:** At time $t_5$, $S_5S_8$ turn off and $D_6D_7$ turn on giving $-V_o$ on the secondary side. On the primary side $S_2S_3$ turn on giving $-V_{in}$, where $V_1 = -V_{in}$ and $V_2 = -nV_o$, resulting in:

$$L\frac{di_L(t)}{dt} = -V_1 + V_2 = -V_{in} + nV_o$$

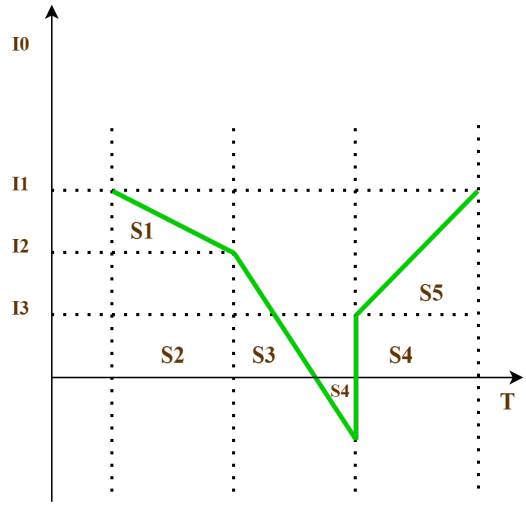



Figure 3.6: Output current in EPS

**From Time Interval I:**

$$i_L(t) = i_L(0) + \frac{(V_{in} + nV_o)t}{L} \quad ; \; 0 \leq t < T_\phi \tag{3.15}$$

**From Time Interval II:**

$$i_L(t) = i_L(T_\phi) + \frac{(V_{in})(t - T_\phi)}{L} \quad ; \; T_\phi \leq t < T_\phi + T_{\phi_o} \tag{3.16}$$

**from Time Interval III:**

$$i_L(t) = i_L(T_\phi + T_{\phi_o}) + \frac{(V_{in} - nV_o))(t - T_\phi - T_{\phi_o})}{L} \quad ; \; T_\phi + T_{\phi_o} \leq t < T_s/2 \tag{3.17}$$

Due to the half-cycle symmetry and with $T_\varphi = \frac{\varphi}{2\pi f_S}$ , $T_{\phi_o} = \frac{\phi_o}{2\pi f_S}$

$$i_L(0) = \frac{\pi(nV_o - V_{dc}) - 2\phi nV_o - \phi_o nV_o}{4\pi f_S L} \tag{3.18}$$

which results for a positive phase shift, $0 < \phi < \pi$ and $0 < \phi_o < \pi$.

The waveform of $I_o$ used for this evaluation is shown in Figure 3.6.

$$I_o = \frac{nV_{in}T_s}{2L}\left[D_2(1-D_2) + \frac{1}{2}D_1(2D_2 - D_1 - 1)\right] \tag{3.19}$$

In the special case where the voltage transmission ratio $k = 1$ and the internal phase shift ratio $D_1 = 0$, the output current reduces to:

$$I_o = \frac{T_s}{2L}D_2(1-D_2)V_{in} \tag{3.20}$$

Finally, the output power $P_o$ is obtained by multiplying the output current with the output voltage $V_o$:

$$P_o = \frac{nV_{in}V_o}{2f_sL}\left[D_2(1-D_2) + \frac{1}{2}D_1(2D_2 - D_1 - 1)\right] \tag{3.21}$$

### 3.2.2 Reactive Power Reduction

In a Dual Active Bridge (DAB) converter, reactive power arises due to the mismatch in polarity between the primary-side voltage $V_p$ and the inductor current $I_L$.

During the interval $t_0$ to $t_1$, switches $S_1$, $S_4$, $S_2$, and $S_3$ are active, resulting in zero voltage across the primary side.As the operation progresses from $t_1$ to $t_2$, the inductor current transitions from negative to positive. In this interval, switches $S_1$ and $S_4$ remain ON, while $S_2$ and $S_3$ are turned OFF, applying a positive voltage $V_p$ across the inductor.

Reactive power is primarily associated with intervals where the inductor current and applied voltage are not aligned, particularly when the current is negative while the voltage is positive. These intervals contribute to non-productive power flow within the converter.

In Single Phase Shift (SPS) control, the elimination of reactive power is only achievable when the voltage conversion ratio satisfies $k = 1$, since only one control variable is available. In contrast, Extended Phase Shift (EPS) control introduces an additional degree of freedom through the inner phase shift parameter $D_1$, enabling better control over reactive power.

The analytical expression for reactive power can be derived by considering the relevant operating intervals, typically mode 2 and mode 6 of the switching cycle.

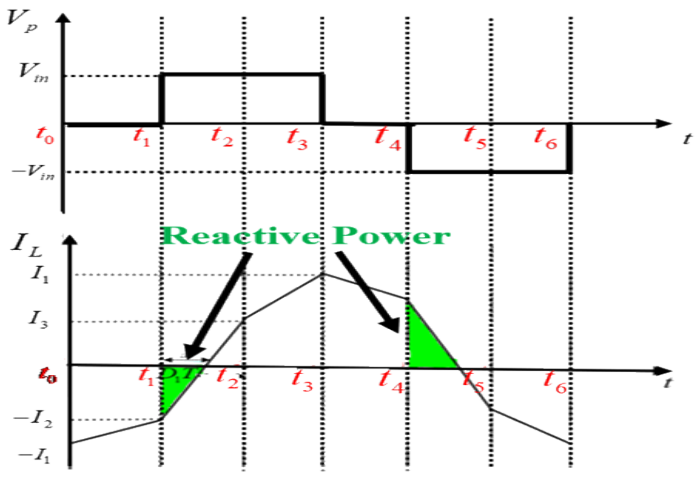



Figure 3.7: Reactive Power

It is observed that $D_2$ is always greater than $D_1$. The reactive power in the DAB converter is governed by three key parameters: the voltage transmission ratio $k$, the internal phase shift ratio $D_1$, and the external phase shift ratio $D_2$.

Reactive power can be minimized by forcing the numerator of the reactive power expression to zero. Based on this condition, the required internal phase shift ratio $D_1$ is given by:

$$D_1 = \frac{2}{2+k} D_2 + \frac{k-1}{k+2} \tag{3.22}$$

For the nominal operating condition, the voltage transmission ratio is set to $k = 1$ Substituting $k = 1$, the relationship between $D_1$ and $D_2$ simplifies to:

$$D_1 = \frac{2}{3} D_2 \tag{3.23}$$

This relationship is useful for controller design, which will be discussed in the subsequent modeling section.

### 3.2.3 Peak Current (Current Stress)

An important design consideration in DAB converters is the current stress experienced by the inductor. The inductor current typically follows a symmetric triangular waveform, governed by the relation:

$$V = L\frac{di}{dt} \tag{3.24}$$

In practical implementations, the leakage inductance of the transformer acts as the main energy transfer element. Therefore, reducing the peak value of the inductor current is essential, as it directly lowers copper losses in the transformer windings and reduces conduction losses in the power switches.

In Extended Phase Shift (EPS) control, the peak current can be effectively controlled by adjusting both the internal phase shift $D_1$ and the external phase shift $D_2$.

The maximum inductor current occurs at the switching instant $t_4$. The analytical expression for the peak current is given by:

$$I_{L,\max} = \frac{T_s n V_o}{4L}\left[k(1-D_1) + (2D_2 - 1)\right] \quad [1] \tag{3.25}$$

From the above expression, it is evident that the peak current in EPS depends on $D_1$ and $D_2$, providing additional flexibility for minimizing current stress compared to conventional SPS control.

**SPS:**

$$i_{pk,N} = \begin{cases} 2(2D - 1 + k), & k \geq 1 \\ 2\left[1 - k(1-2D)\right], & k < 1 \end{cases} \quad [1] \tag{3.26}$$

**EPS:**

$$i_{pk,N} = \begin{cases} 2\left[k(1-D_1) + (2D_1 + 2D_2 - 1)\right], & k \geq 1 \\ 2\left[1 - k(1-D_1-2D_2)\right], & k < 1 \end{cases} \quad [1] \tag{3.27}$$

.

Due to the introduction of the inner phase shift $D_1$, the peak current (current stress) in EPS is reduced compared to SPS. In particular, EPS exhibits lower current stress than SPS for $k \geq \sqrt{2}$.

### 3.2.4 ZVS Analysis in EPS

Zero Voltage Switching (ZVS) is achieved when the inductor current has the appropriate polarity at the switching instant, allowing the switch capacitance to discharge to zero before turn-on.

For EPS control, the ZVS conditions are as follows:

- For primary-side switches ($S_1$, $S_3$):

$$i_L(t_0) \leq 0$$

- For secondary-side switches ($S_5$):

$$i_L(t_s) \geq 0$$

The ZVS condition corresponding to $i_L(t_0) \leq 0$ can be expressed as:

$$k(1 - D_1) + (2D_1 + 2D_2 - 1) \geq 0 \tag{3.28}$$

Compared to SPS, EPS significantly expands the ZVS operating region, particularly under light-load conditions and when the voltage conversion ratio deviates from unity.

In particular, during Mode 2 operation, all switches can achieve ZVS, making it the preferred mode for optimal performance.

# Chapter 4

# Simulation Observations and Analysis

## 4.1 DAB operation in Battery to Grid(B2G) and Grid to Battery condition(G2B)

### 4.1.1 Control Scheme of DAB

The control strategy of the Dual Active Bridge (DAB) converter is based on phase-shift modulation between the primary and secondary bridges.A proportional-integral (PI) controller is used to regulate the reference current.

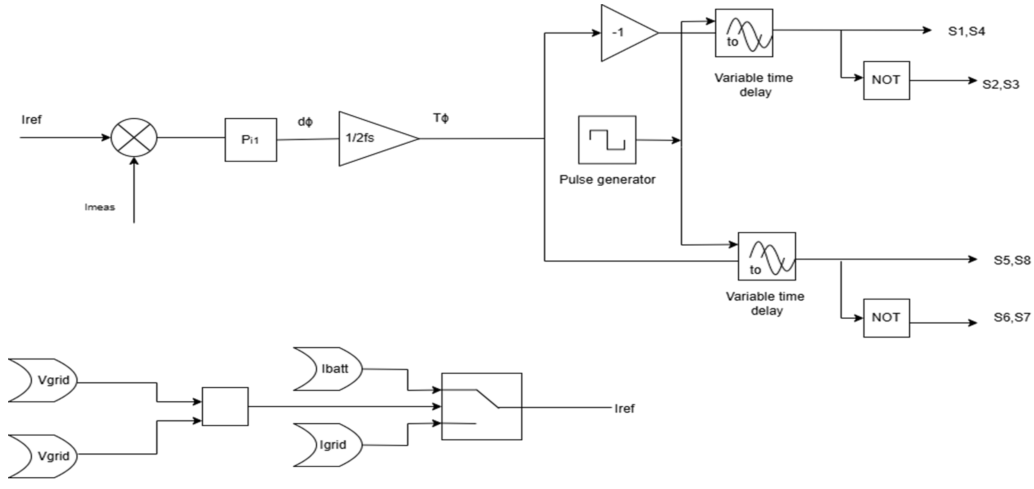



Figure 4.1: Control scheme of single-phase DAB converter

The error between reference current ($I_{ref}$) and measured current is processed through

the PI controller to generate the phase shift ($\phi$). This phase shift controls the power flow between the two ports.

If $V_{batt} > V_{grid}$, the phase shift is positive, resulting in power transfer from battery to grid.

If $V_{batt} < V_{grid}$, the phase shift becomes negative, enabling reverse power flow from grid to battery.

### 4.1.2 Simulation Parameters

Table 4.1: Simulation parameters of the DAB converter

| Parameter | Value |
|---|---|
| Input Voltage ($V_1$) | 800 V |
| Output Voltage ($V_2$) | 380–500 V |
| Switching Frequency ($f_s$) | 100 kHz |
| Transformer Ratio ($n$) | 1.6 |
| Phase Shift ($\phi$) | 0.46–0.65 rad |
| Power ($P$) | 10 kW |
| Inductance ($L$) | 40 $\mu$H |
| Resistance ($R$) | 14–25 $\Omega$ |
| Capacitance ($C$) | 9 $\mu$F |

### 4.1.3 Case Study 1: Power Transfer from Battery to Grid

In this case, the reference current is set to $I_{ref} = 530$ A. The phase shift is positive ($\phi = 54°$), resulting in power transfer from the battery to the grid.

- Input voltage ($V_{batt}$): 745 V
- Output voltage ($V_{grid}$): 755 V
- Battery current ($I_{batt}$): $\approx$ 530 A
- Grid current ($I_{grid}$): 517 A

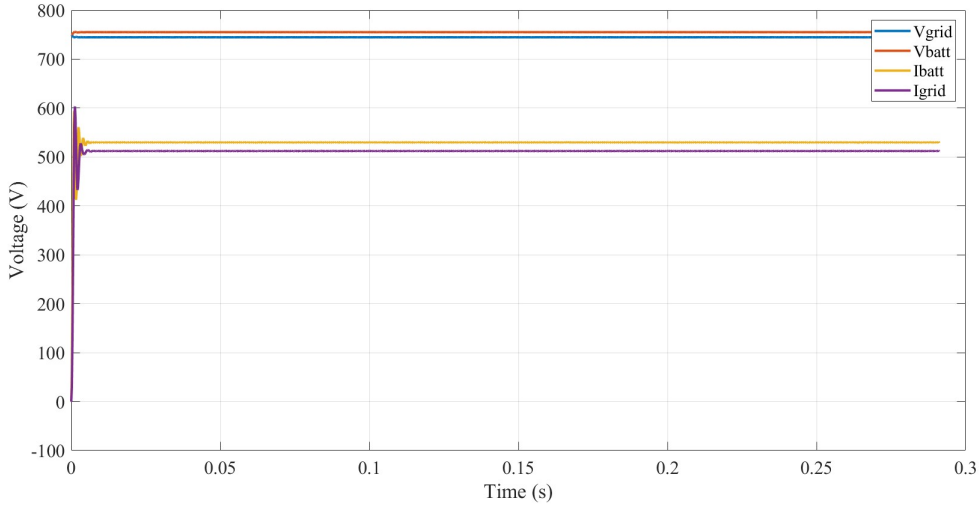



Figure 4.2: Simulation results for $\phi = 54°$ showing power flow from battery to grid

It is observed that the battery current is positive, indicating discharge, while the grid current follows the reference value. The phase shift control successfully regulates the power flow.

### 4.1.4 Case Study 2: Power Transfer from Grid to Battery

In this case, the reference current is set to $I_{ref} = -530$ A. The phase shift becomes negative ($\phi = -54°$), enabling power transfer from the grid to the battery.

- Input voltage ($V_{batt}$): 755 V
- Output voltage ($V_{grid}$): 745 V
- Battery current ($I_{batt}$): $\approx -530$ A
- Grid current ($I_{grid}$): $\approx -530$ A

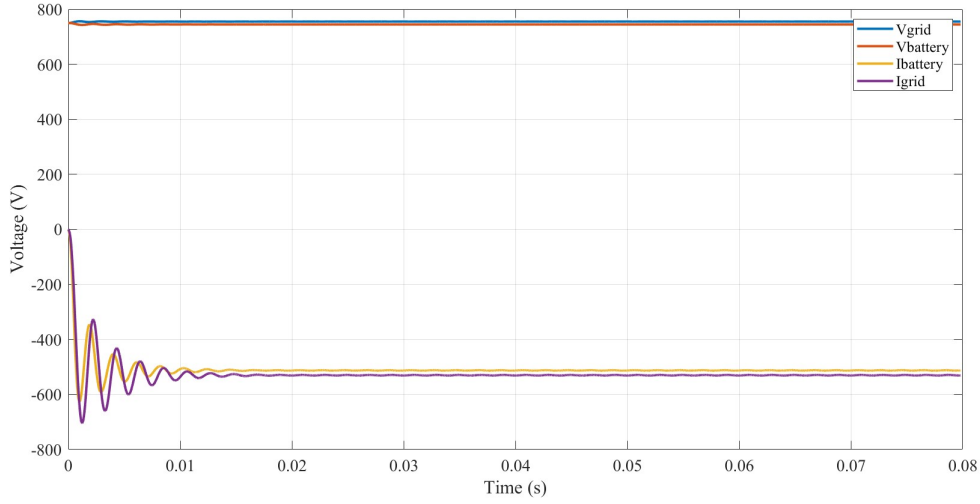



Figure 4.3: Simulation results for $\phi = -54°$ showing power flow from grid to battery

The negative current confirms charging operation. The control system effectively reverses power flow by adjusting the phase shift.

**Discussion**: The simulation results confirm that the phase-shift control of the DAB converter enables bidirectional power flow.A positive phase shift results in power transfer from battery to grid, whereas a negative phase shift reverses the direction.The PI controller ensures accurate tracking of the reference current under both operating conditions.

# Chapter 5

# Variable Transformer Equivalent Turns Ratio Dual-Active Bridge Converter for Wide Voltage Range Applications

## 5.1 Variable Transformer DAB Converter

The topology structure of VTDAB converter is illustrated in Figure 4. VTDAB converter is an advanced modification of the conventional DAB topology designed for wide voltage range applications. VTDAB aims to decrease inductor rms current and increase the efficiency of the Conventional DAB (CDAB) for wide voltage range applications.

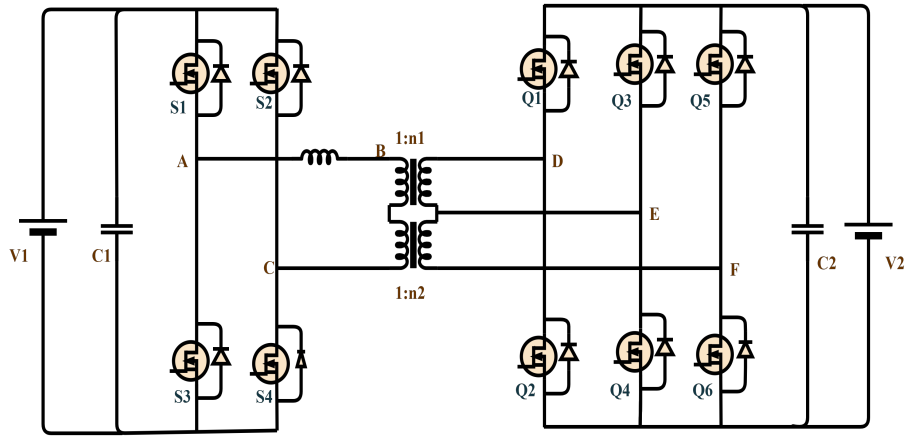



Figure 5.1: Variable Transformer Dual Active Bridge Converter

## 5.2 Advantages of VTDAB over CDAB:

Under ideal conditions the optimal voltage gain $G$ equals the turn's ratio when the actual voltage gain $G$ deviates significantly from this fixed ratio the efficiency drops and performance deteriorates because:

- The inductor RMS current increases.
- The converter may lose ZVS.
- Switching and conduction losses increase.
- Reactive power and circulating current rise

Depending on different driving rules for secondary-side switches dual transformer realizes three different equivalent turns ratios so it has 3 operating modes. The detailed analysis of each mode and the key waveforms of VTDAB under different operating modes during a cycle are shown in Figure 5.2, TPS modulation is adopted for the VTDAB.

## 5.3 Optimal Design and Control

Optimal phase-shift control (OPS) proposed which has the characteristics of simplicity, high efficiency, and applicability to different operating conditions. The OPS adopts the TPS modulation and can optimize $IL_{rms}$ of the CDAB.

The normalized power is expressed as:

$$p^* = \frac{P_{\text{load}}}{P_{\text{max}}} \tag{5.1}$$

The complete control scheme for VTDAB is shown in Figure 4.3. This control method samples the $V2, V1$ and secondary current $i2$, and calculates voltage gain $G$ and normalized power $p$ of the converter. The mode of the VTDAB is selected and the values of $neq$,$M$,and $p*$ are updated before entering the control algorithm.

Therefore, the three operating modes of the VTDAB can be switched according to the following equations:

$$\text{Mode A} : G \leq 0.8752p^3 - 0.4966p^2 + 0.0799p \quad [2] \tag{5.2}$$

$$\begin{aligned}\text{Mode C} : G &< 0.8752p^3 - 0.4966p^2 + 0.0799p \;\&\& \\ G &> 8.191p^3 - 3.566p^2 + 0.4859p + 1.203 \quad [2]\end{aligned} \tag{5.3}$$

$$\text{Mode B} : G \geq 8.191p^3 - 3.566p^2 + 0.4859p + 1.203 \quad [2] \tag{5.4}$$

**Mode A:** In Mode A, switches Q3 and Q5, and Q4 and Q6 operate simultaneously, resulting in transformer T2 being short-circuited. Hence, only transformer T1 participates in power transfer. The equivalent turns ratio is:

$$n_{eq} = n_1$$

**Mode B:** In Mode B, switches Q1 and Q3, and Q2 and Q4 operate simultaneously, which short-circuits transformer T1. Therefore, only transformer T2 is active. The equivalent turns ratio is:

$$n_{eq} = n_2$$

**Mode C:** In Mode C, switches Q1 and Q5, and Q2 and Q6 operate simultaneously. In this mode, both transformers T1 and T2 are active and operate with opposite polarity. The equivalent turns ratio is:

$$n_{eq} = \frac{n_1 n_2}{n_2 - n_1}$$

Now based on OPS control shown in Table 1 the values for each phase-shift ratio are generated, the control scheme produces the corresponding PWM driving signals based on the operating mode and the magnitude of each phase shift ratio.

| **Voltage Ratio (M)** | **Power Range ($p^*$)** | **Optimal Control Parameters** |
|---|---|---|
| $M < 0.98$ | $0 \le p^* < 2M(1-M)$ | $D_1 = 1 - \sqrt{Mp^*/(2-2M)}$<br>$D_2 = 1 - \sqrt{p^*/(2M-2M^2)}$<br>$D_0 = D_1 - D_2$ |
| $M < 0.98$ | $2M(1-M) \le p^* \le 1$ | $D_1 = (1-M)\sqrt{(1-p^*)/(1-2M+2M^2)}$<br>$D_2 = 0$<br>$D_0 = (1-D_1-M)/(2(1-M)) + D_1$ |
| $M > 1.02$ | $0 \le p^* \le (2-2M)/M^2$ | $D_2 = 1 - \sqrt{p^*/(2M-2)}$<br>$D_1 = 1 - M(1-D_2)$<br>$D_0 = 0$ |
| $M > 1.02$ | $(2-2M)/M^2 \le p^* \le 1$ | $D_2 = (M-1)\sqrt{(1-p^*)/(M^2-2M+2)}$<br>$D_1 = 0$<br>$D_0 = ((2-M)(1-D_2)+2M-3)/(2(M-1)) - D_2$ |
| $0.98 \le M \le 1.02$ | $0 \le p^* \le 1$ | $D_1 = 0$<br>$D_2 = 0$<br>$D_0 = 1 - \sqrt{(1-p^*)}/2$ |

Table 5.1: Optimal Phase Shift Control [2]

For optimal performance, the voltage ratios of the three modes $M_a$, $M_b$, and $M_c$ should be near unity and close to each other:

$$M_a \approx M_b \approx M_c \;\Rightarrow\; \frac{V_{2,a}}{n_{eq,a}V_1} \approx \frac{V_{2,b}}{n_{eq,b}V_1} \approx \frac{V_{2,c}}{n_{eq,c}V_1} \tag{5.5}$$

To achieve this, the turns ratios should maintain an equal-ratio relationship:

$$\frac{n_1}{n_2 - n_1} = \frac{n_2}{n_2 - n_1} = 3 \;\Rightarrow\; \text{Solve to find } n_1, n_2 \tag{5.6}$$

After evaluating several cases, the optimal values selected are:

$$n_1 = 0.556, \qquad n_2 = 1.5 \tag{5.7}$$

as they provide the lowest $i_{L,\text{rms}}$ across the full voltage range from 0.4 to 2.5.

Table 5.2: Optimal Designed Turns Case

| $T_1$ ($n_1$) | $T_2$ ($n_2$) | $n_{eq,a}$ | $n_{eq,c}$ | $n_{eq,b}$ |
|---|---|---|---|---|
| 0.556 | 1.5 | 0.556 | 0.882 | 1.5 |

To achieve optimal power transfer with minimum inductor RMS current, the phase shifts must be determined according to the following constraints:

1. Ensure correct power transfer using the power equation:

$$P = \frac{nV_1V_2\,\phi(\pi - |\phi|)}{2\pi^2 f_s L_s} \tag{5.8}$$

2. Minimize:

$$i_{L,\text{rms}} = \sqrt{\sum_{k=1,3,5,\ldots} \left[\frac{8\sqrt{2}}{\pi^2 k^2 G}\sqrt{A(k)^2 + B(k)^2}\right]^2} \tag{5.9}$$

where $A(k)$ and $B(k)$ are derived from the voltage harmonics of $v_{AC}$ and $v_{BC}$.

3. Maintain ZVS by ensuring proper current polarity during dead-time.

Using Lagrangian optimization and solving for conditions that minimize current while achieving the desired power transfer, the optimal values of $D_0$, $D_1$, and $D_2$ are determined for different regions of $M$ and $p^*$.

### 5.3.1 Mathematical Model

By applying Fourier series decomposition, the square-wave voltages $V_{AC}$ and $V_{BC}$ can be expressed in the frequency domain, enabling harmonic analysis and facilitating the derivation of power transfer relationships.

$$V_{AC}(t) = \sum_{k=1,3,5,\ldots}^{\infty} \frac{4V_1}{k\pi} \cos\left(\frac{kD_1\pi}{2}\right) \sin\left(k\omega t - \frac{kD_1\pi}{2}\right) \tag{5.10}$$

$$V_{BC}(t) = \sum_{k=1,3,5,\ldots}^{\infty} \frac{4V_1}{n_{eq}k\pi} \cos\left(\frac{kD_2\pi}{2}\right) \sin\left(k\omega t - kD_0\pi - \frac{kD_2\pi}{2}\right) \tag{5.11}$$

where $k$ is the harmonic order, $f$ is the switching frequency, and $\omega = 2\pi f$.

The inductor current $i_L(t)$ in the DAB converter can be expressed as:

$$i_L(t) - i_L(t_0) = \frac{1}{L} \int_{t_0}^{t} \left[V_{AC}(\tau) - V_{BC}(\tau)\right] d\tau \tag{5.12}$$

Based on the symmetry of $i_L(t)$, we have:

$$i_L\left(\frac{T}{2}\right) = i_L\left(-\frac{T}{2}\right) = -i_L(t_0)$$

Solving the integral yields the current expression:

$$i_L(t) = \sum_{k=1,3,5,\ldots}^{\infty} \frac{4\omega\pi}{kL} \sqrt{A(k)^2 + B(k)^2} \sin\left(k\omega t + \arctan\left(\frac{A(k)}{B(k)}\right)\right) \tag{5.13}$$

The coefficients $A(k)$ and $B(k)$ are given by:

$$A(k) = \frac{G}{2n_{eq}} \left[\cos(k\pi D_0) + \cos(k\pi D_2) + \cos(k\pi D_0) + \cos(k\pi D_1)\right] \tag{5.14}$$

$$B(k) = \frac{G}{2n_{eq}} \left[\sin(k\pi D_0) + \sin(k\pi D_2) + \sin(k\pi D_0) + \sin(k\pi D_1)\right] \tag{5.15}$$

where $G$ is the voltage gain of the converter, defined as:

$$G = \frac{V_2}{V_1} \tag{5.16}$$

The transmission power $P$ of the DAB converter is given by:

$$P = \int_0^T V_{AC}(t)\, i_L(t)\, dt \tag{5.17}$$

$$p = \sum_{k=1,3,5,\ldots} \frac{32 \cos\left(\frac{kD_1\pi}{2}\right)}{\pi^3 k^3 G} \left[\cos\left(\frac{kD_1\pi}{2}\right) B(k) - \sin\left(\frac{kD_1\pi}{2}\right) A(k)\right] \tag{5.18}$$

$$i_{L,\text{rms}} = \sqrt{\sum_{k=1,3,5,\ldots} \left[\frac{8\sqrt{2}}{\pi^2 k^2 G} \sqrt{A(k)^2 + B(k)^2}\right]^2} \tag{5.19}$$

### 5.3.2 ZVS Conditions

The currents through the bridge legs on the secondary side of the converter, denoted as $i_{q12}$, $i_{q34}$, and $i_{q56}$, are essential in ensuring Zero Voltage Switching (ZVS) of the secondary-side switches. These currents are determined by the transformer-coupled inductor current $i_L$ and the transformer turns ratios $n_1$ and $n_2$, and are expressed as:

$$i_{q12}(t) = \frac{i_L(t)}{n_1} \tag{5.20}$$

$$i_{q56}(t) = \frac{i_L(t)}{n_2} \tag{5.21}$$

$$i_{q34}(t) = -\, i_{q12}(t) + i_{q56}(t) \tag{5.22}$$

The direction of $i_{q12}$ is the same as that of $i_L$, while $i_{q34}$ and $i_{q56}$ flow in the opposite direction relative to $i_L$ and $i_{q12}$.

The ZVS condition is satisfied when the current at the switching instant is sufficient to fully discharge the MOSFET's output capacitance before the gate signal turns it ON. The tables below summarize the necessary ZVS conditions for both the primary and secondary-side switches in VTDAB.

Table 5.3: ZVS Conditions for Primary-Side Switches (VTDAB)

| Switch | ZVS Condition |
|---|---|
| $S_1$ | $i_L(t_0) < -2i_{c_s}$ |
| $S_2$ | $i_L(t_0 + T_h) > 2i_{c_s}$ |
| $S_3$ | $i_L(t_2 + T_h) > 2i_{c_s}$ |
| $S_4$ | $i_L(t_2) < -2i_{c_s}$ |

Table 5.4: ZVS Conditions for Secondary-Side Switches (Mode A)

| Switch | ZVS Condition |
|---|---|
| $Q_1$ | $i_{q12}(t_4) > 2i_{c_q}$ |
| $Q_2$ | $i_{q12}(t_4 + T_h) < -2i_{c_q}$ |
| $Q_3$ | $i_{q34}(t_6 + T_h) > 2i_{c_q}$ |
| $Q_4$ | $i_{q34}(t_6) < -2i_{c_q}$ |
| $Q_5$ | $i_{q56}(t_6 + T_h) > 2i_{c_q}$ |
| $Q_6$ | $i_{q56}(t_6) < -2i_{c_q}$ |

where $i_{c_s}$ is the primary-side capacitor current, $i_{c_q}$ is the secondary-side capacitor current, and $i_L$ is the inductor current.

## 5.4 Comparative Simulation Analysis of SPS, EPS, DPS, and TPS-Based VTDAB Converter

This section presents a comparative analysis of different modulation techniques, namely SPS,EPS,DPS,TPS applied to the Conventional Dual Active Bridge (CDAB) and the proposed Variable Transformer Dual Active Bridge (VTDAB) converter.

### Simulation Parameters

The simulations are carried out under identical operating conditions to ensure a fair comparison. The parameters used are listed below:

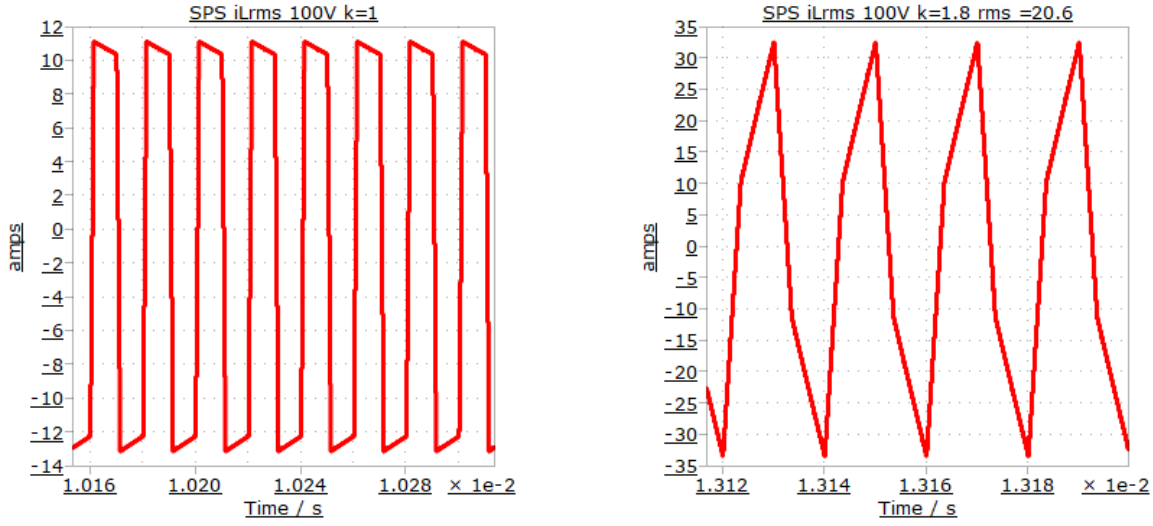
Table 5.5: Simulation Parameters for Comparative Analysis

| Parameter | Value |
|---|---|
| Input Voltage ($V_1$) | 100 V |
| Output Voltage Range ($V_2$) | 40 V – 250 V |
| Switching Frequency ($f_s$) | 50 kHz |
| Rated Power ($P$) | 1 kW |
| Transformer Turns Ratio ($n$) | 1 |

To analyze wide voltage operation, the output voltage is varied while keeping $V_1$ and $n$ constant. The results are evaluated at two operating points:

- $V_2 = 100$ V ($k = 1$)
- $V_2 = 55$ V ($k = 1.8$)

**Simulation Results of SPS**




(a) $V_2 = 100$ V (b) $V_2 = 55$ V

Figure 5.4: Simulation results of SPS modulation under different voltage conditions

Under SPS control, significant circulating current is observed when the voltage ratio deviates from unity. This leads to higher inductor RMS current and reduced efficiency.

### Simulation Results of EPS

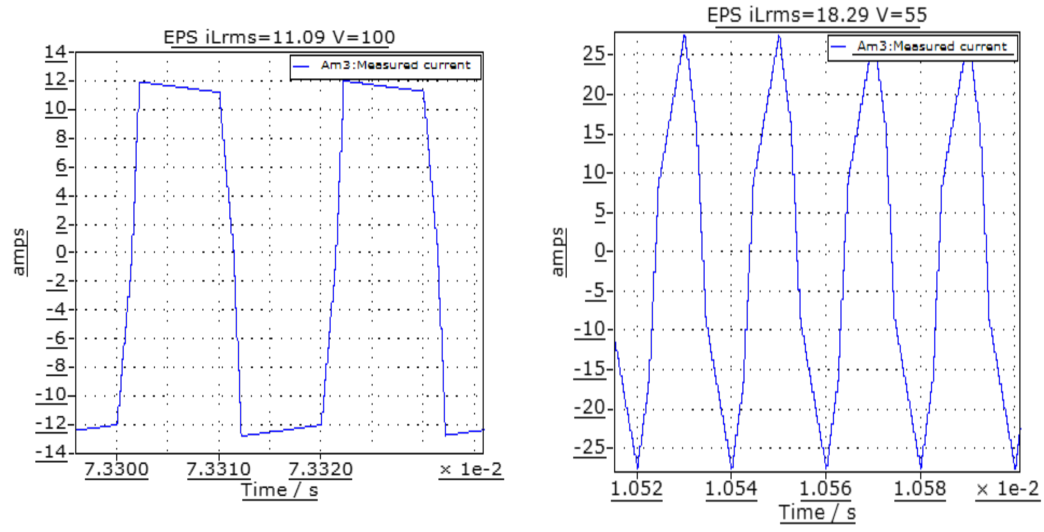



Figure 5.5: Simulation results of EPS modulation

EPS modulation introduces an additional degree of freedom, which helps reduce circulating current compared to SPS. However, under wide voltage variation, current stress is still significant.

### Simulation Results of DPS

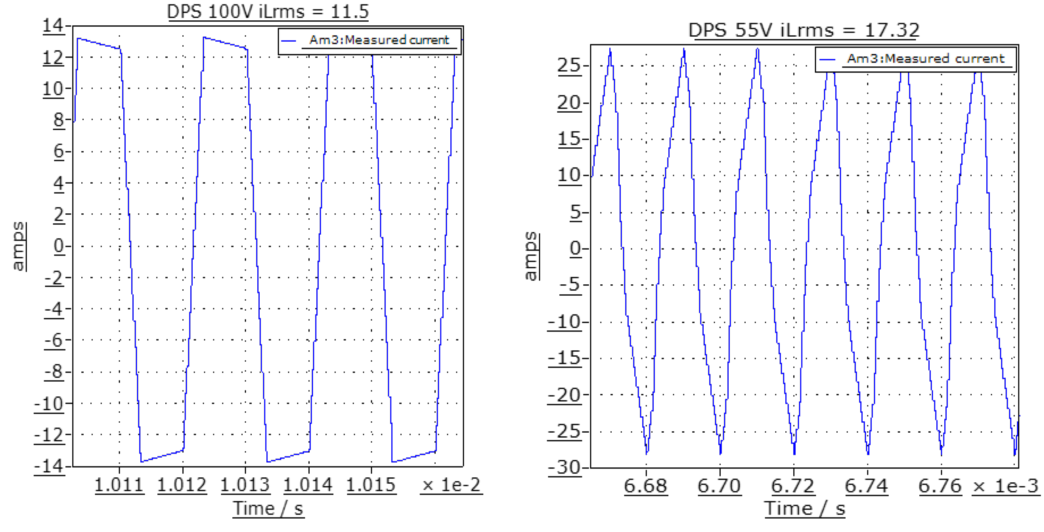



Figure 5.6: Simulation results of DPS modulation

DPS further improves current shaping and reduces RMS current compared to SPS and EPS. Nevertheless, performance degradation is still observed under large voltage mismatches.

**Simulation Results of TPS-Based VTDAB**

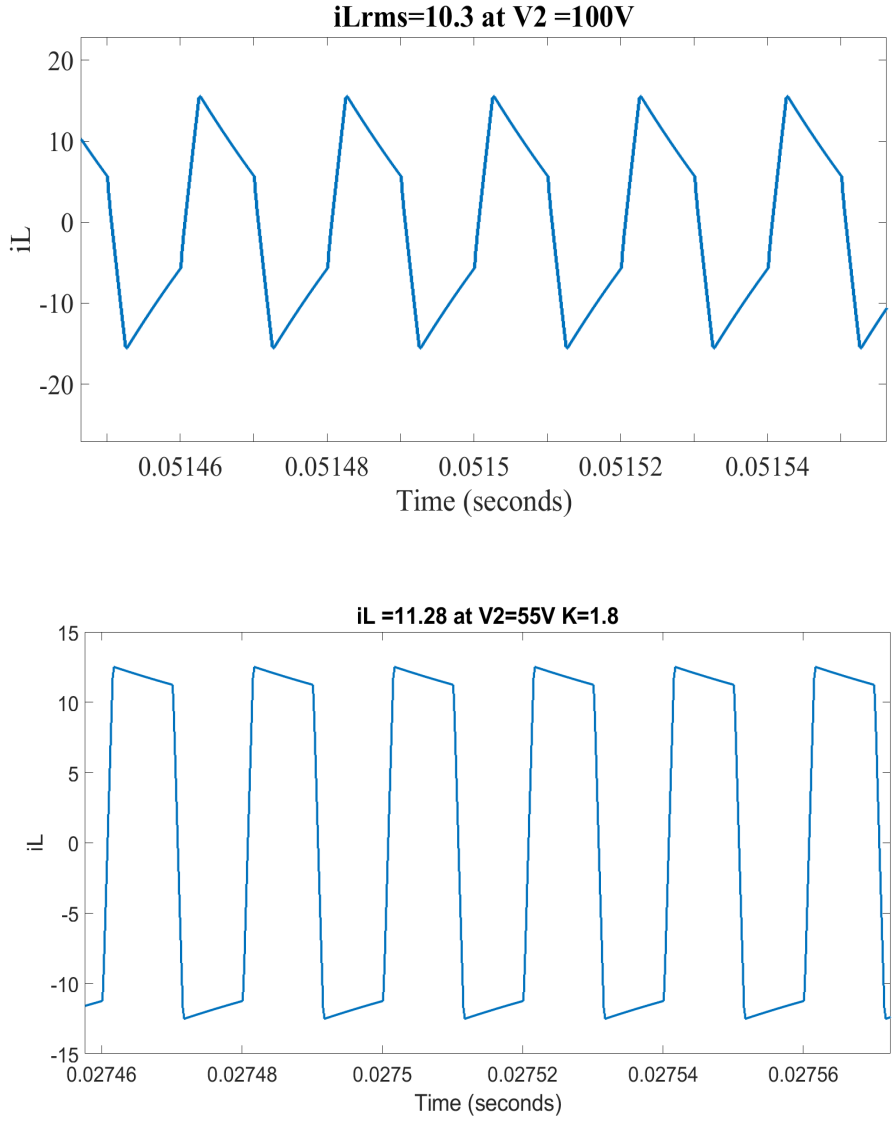
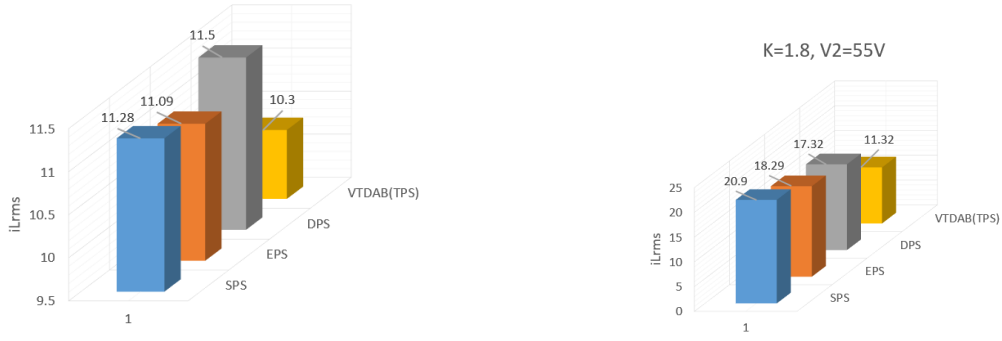



Figure 5.7: Simulation results of TPS-based VTDAB converter

The proposed VTDAB converter employing TPS modulation helps reduces circulating current and achieves lower inductor RMS current across the entire voltage range.

### Inductor RMS Current Comparison




(a) $V_2$ = 100 V (b) $V_2$ = 55 V

Figure 5.8: Inductor RMS Current Comparison

### Discussion

From the comparative analysis, it is evident that:

- SPS exhibits the highest circulating current under voltage mismatch conditions.
- EPS and DPS provide partial improvement but still suffer under wide voltage range operation.
- The proposed VTDAB with TPS control achieves the lowest RMS current and maintains efficient operation.

The reduction in RMS current directly contributes to lower conduction losses, improved efficiency, and enhanced soft-switching capability.

Furthermore, the proposed topology demonstrates robust operation over a wide voltage range of 40 V to 250 V, making it suitable for applications such as EV charging systems and battery energy storage systems.

Simulation results also indicate that the converter can tolerate up to 30–40% deviation between $V_2$ and $nV_1$ while maintaining Zero Voltage Switching (ZVS) and reduced current stress.

The proposed TPS-based VTDAB converter outperforms conventional SPS, EPS, and DPS-controlled DAB converters in terms of RMS current reduction and efficiency under wide voltage operating conditions.

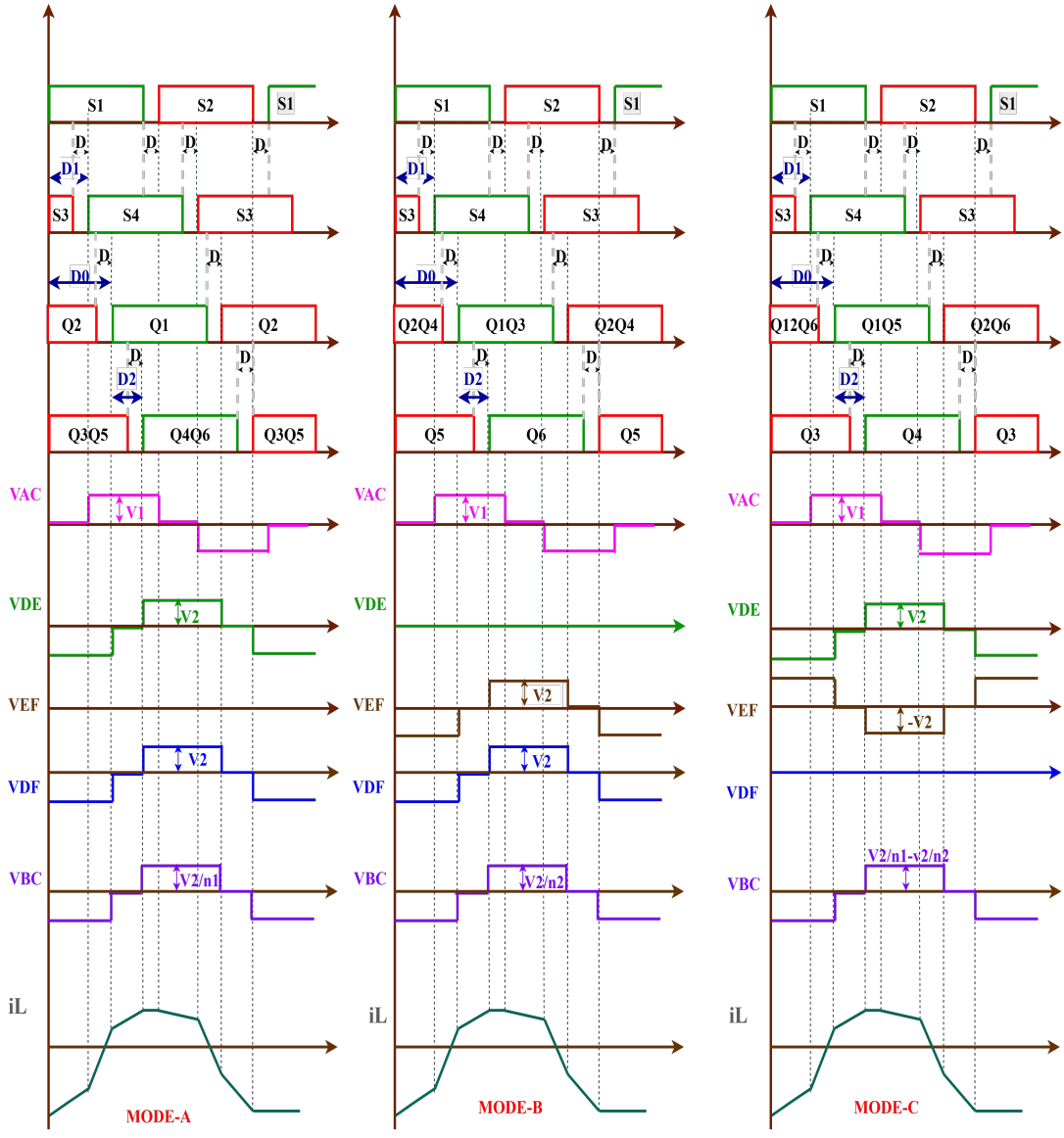



Figure 5.2: Key waveforms of VTDAB (a) Mode A. (b) Mode B. (c) Mode C

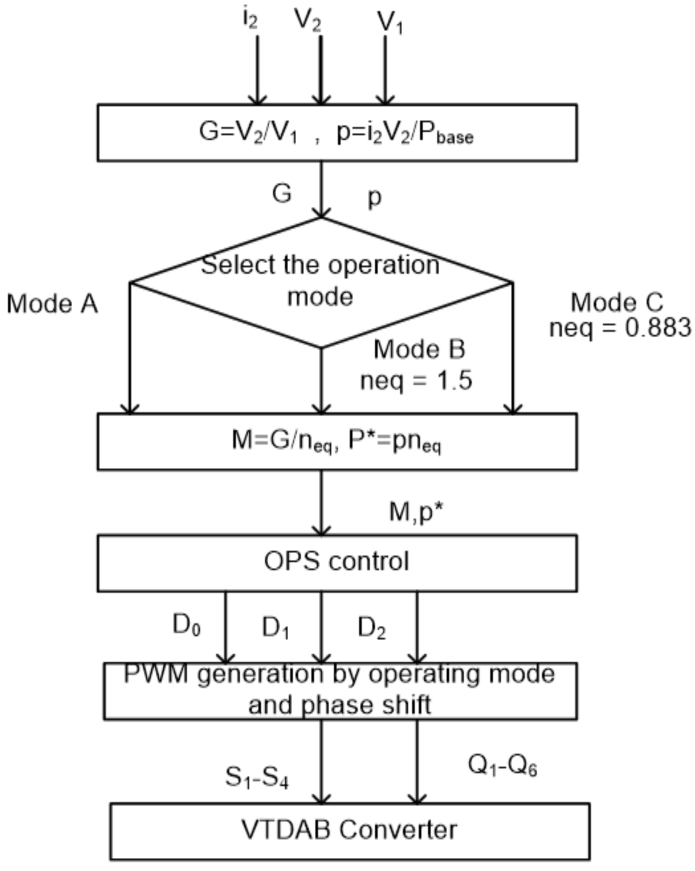



Figure 5.3: Complete control algorithm of VTDAB

# Chapter 6

# Hardware Design of VTDAB Converter

## 6.1 Introduction

This chapter presents the detailed hardware design of the proposed Variable Transformer Dual Active Bridge (VTDAB) converter. The design focuses on achieving high efficiency, wide voltage range operation, and reliable bidirectional power transfer.

## 6.2 Hardware specifications

Table 6.1: System Specifications

| Parameter | Value |
|---|---|
| Input Voltage ($V_1$) | 100 V |
| Output Voltage ($V_2$) | 40–250 V |
| Rated Power | 1 kW |
| Switching Frequency ($f_s$) | 50 kHz |
| Topology | VTDAB |
| Transformer Type | Dual Transformer |

## 6.3 Component Selection for Power Card

### 6.3.1 MOSFET Selection Criteria

The selection of the power semiconductor device is critical for achieving high efficiency, reliability, and stable operation in high-frequency converters.The MOSFET must be selected based on electrical stress, switching performance, thermal limits, and loss minimization.

In this work, the SiC MOSFET C3M0025065D is selected after evaluating key parameters as discussed below.

#### Voltage Rating ($V_{DS}$)

The drain-to-source breakdown voltage $V_{DS(\max)}$ must exceed the maximum voltage stress across the switch, including transient overshoots caused by parasitic inductance and LC ringing.

Due to high-frequency switching and leakage inductance effects, voltage overshoot is significant. A safety margin of 1.5–2 times the operating voltage is typically considered.

$$V_{DS(\max)} \geq 2 \times V_{\text{operating}} \tag{6.1}$$

For the given system:

Maximum DC bus voltage $\approx 250\,\text{V}$
Selected device: 650 V rating

This enables sufficient margin for transient spikes and ensures safe operation.

#### Gate Voltage Requirements ($V_{GS}$)

The maximum gate-to-source voltage must not exceed device limits, and the operating gate voltage $V_{GS(\text{op})}$ must be chosen from the transfer characteristics.

- Ensure $V_{GS(\max)} > V_{\text{driver}}$
- Select $V_{GS(\text{op})}$ slightly above the $R_{DS(\text{on})}$ knee point

Proper gate voltage ensures:

- Minimum conduction losses
- Reliable switching
- Avoidance of device degradation

### Current Rating ($I_D$ and $I_{DM}$)

The MOSFET must handle both continuous and peak currents.

**Continuous current:**

$$I_D \geq 1.5 \times I_{\text{load}} \tag{6.2}$$

**Peak (pulsed) current:**

$$I_{DM} \geq I_{\text{peak}} \tag{6.3}$$

For the system:

$$I_{\text{load}} = \frac{1000}{40} = 25\,\text{A}$$

Design margin $\rightarrow$ 50 A required

Selected MOSFET:

$$I_D = 70\,\text{A} \quad \text{(Safe)}$$

High pulsed current capability ensures handling of transients.

### On-State Resistance ($R_{DS(\text{on})}$)

When the MOSFET is ON, it behaves like a resistor. Conduction loss is given by:

$$P_{\text{cond}} = I_{\text{rms}}^2 \cdot R_{DS(\text{on})} \tag{6.4}$$

Lower $R_{DS(\text{on})}$ reduces conduction losses but increases device capacitances (especially $C_{\text{oss}}$), leading to higher switching losses. Therefore, a trade-off is required.

Selected device:

$$R_{DS(\text{on})} = 25\,\text{m}\Omega \quad \text{(typical)}$$

**Switching Loss Consideration**

Switching losses are given by:

$$P_{\text{sw}} = f_s \cdot (E_{\text{on}} + E_{\text{off}}) \tag{6.5}$$

For high-frequency operation, minimizing switching energy is critical. SiC MOSFETs inherently offer:

- Faster switching
- Lower switching losses
- Reduced overlap between voltage and current

**Gate Charge ($Q_g$)**

Gate charge determines how quickly the MOSFET can be switched.

- Lower $Q_g \rightarrow$ faster switching
- Lower driver losses

Selected device:

$$Q_g = 108\,\text{nC}$$

### Capacitances ($C_{iss}$, $C_{oss}$, $C_{rss}$)

MOSFET parasitic capacitances significantly influence switching behavior:

- $C_{iss}$: affects gate drive requirement
- $C_{oss}$: affects turn-off behavior and ZVS condition
- $C_{rss}$ (Miller capacitance): affects switching speed and stability

High $C_{oss}$ leads to:

- Increased switching loss
- LC ringing

However, sufficient $C_{oss}$ can assist in achieving ZVS in DAB converters. Hence, selection must balance switching loss and soft-switching requirements.

### Reverse Recovery and Body Diode Performance

In conventional silicon MOSFETs, reverse recovery charge $Q_{rr}$ contributes to significant switching losses and EMI.

SiC MOSFET advantages:

- Negligible reverse recovery
- Faster diode recovery
- Reduced current spikes

### Switching Timing Parameters

Fast switching reduces losses but increases EMI. Key parameters include:

- Turn-on delay $T_d(\text{on})$

- Rise time $T_r$
- Turn-off delay $T_d(\text{off})$
- Fall time $T_f$

The selected device exhibits fast transition times, reducing overlap losses.

### Thermal Considerations

Thermal performance is critical for reliability.

$$T_j = T_a + P_{\text{loss}} \cdot R_{\theta JA} \tag{6.6}$$

where:

- $T_j$: junction temperature
- $R_{\theta JA}$: thermal resistance

Selected device:

$$T_j(\text{max}) = 175^\circ\text{C}, \quad P_D = 326\,\text{W}$$

This ensures safe operation under high-power conditions with proper heat sinking.

### Final Selection Justification

The C3M0025065D SiC MOSFET is selected due to:

- High voltage rating (650 V) ensuring safe operation under transients
- Low $R_{DS(\text{on})}$ reducing conduction losses
- Low gate charge enabling high-speed switching
- Minimal reverse recovery improving efficiency

- High thermal capability (175°C operation)
- Suitability for high-frequency, high-efficiency DAB converters

Compared to silicon MOSFETs and IGBTs, SiC devices offer superior performance in terms of switching speed, efficiency, and thermal robustness, making them ideal for this application.

### 6.3.2 Input/Output Capacitors (450LXW220MEFR18X50)

In power electronic circuits like switch-mode power supplies (SMPS), DC-DC converters, inverters, and rectifiers, input and output capacitors play essential roles in maintaining stable operation and improving performance. The 450LXW220MEFR18X50 capacitor is a highly reliable component designed for high-voltage, high-current applications. Its balanced specifications make it ideal for industrial power supplies, energy storage, and high-power converters. The low ESR and high ripple current capabilities ensure effective performance in demanding environments.

**General Specifications**

Table 6.2: General Specifications of 450LXW220MEFR18X50

| **Parameter** | **Value** |
|---|---|
| Part Number | 450LXW220MEFR18X50 |
| Series | LXW |
| Capacitance | 220 $\mu$F |
| Voltage Rating | 450 V DC |
| Tolerance | ±20% |
| Temperature Range | −40 °C to +105 °C |
| Case Size | 18 mm (D) × 50 mm (H) |
| Terminal Type | Radial, Snap-In |
| Ripple Current (100 kHz) | 1.23 A |
| ESR (Typical) | Low |
| Lifetime | 12000 hours at 105 °C |
| Mounting | PCB through-hole |

#### Capacitance and Voltage Rating

The capacitance of 220 $\mu$F and a voltage rating of 450 V make it suitable for applications requiring significant energy storage and voltage stabilization. The tolerance of ±20% is typical for electrolytic capacitors.

#### ESR and Ripple Current

- The Equivalent Series Resistance (ESR) is low, minimizing power loss.
- The high ripple current capability of 1.23 A ensures stable performance in high-frequency switching environments.

#### Thermal Characteristics

Operating temperature range: −25 °C to +105 °C with a life expectancy of 12000 hours at 105 °C. At lower temperatures, the lifespan increases.

### 6.3.3 Decoupling Capacitors Across Each Leg

High-frequency decoupling capacitors are placed across the DC bus of each MOSFET leg to ensure stable and reliable switching operation. In this work, multiple ceramic capacitors (Part No: CGA5L4X7T2W104M160AA) are connected in parallel and positioned as close as possible to the MOSFET terminals to minimize parasitic effects.

During fast switching transitions, the current exhibits a high rate of change ($di/dt$), which can lead to voltage spikes due to parasitic inductance in PCB traces and device packaging. These spikes are governed by:

$$V = L\frac{di}{dt} \tag{6.7}$$

Additionally, the interaction between stray inductance and MOSFET output capacitance ($C_{\text{oss}}$) results in LC ringing. The stored energy in inductance is given by:

$$E = \frac{1}{2}Li^2 \tag{6.8}$$

This energy is exchanged with parasitic capacitances, causing oscillations that contribute to electromagnetic interference (EMI), noise in measurement circuits, and potential device stress.

Decoupling capacitors mitigate these issues by providing a local low-impedance current path, thereby suppressing voltage spikes, reducing switching ripple, and preventing high-frequency disturbances from propagating to the source.

### Purpose of Parallel Capacitors

Placing capacitors in parallel across each leg enables the following advantages:

- **Increased Effective Capacitance:**

$$C_{\text{total}} = C_1 + C_2 + \cdots + C_n$$

- **Voltage Stabilization:** Reduces DC bus voltage ripple during switching transients.
- **High-Frequency Noise Suppression:** Effectively filters switching noise and reduces EMI.
- **Reduced ESR and ESL:** Parallel combination lowers equivalent series resistance (ESR) and inductance (ESL), improving transient response and reducing voltage overshoot.
- **Improved Current Sharing:** Distributes ripple current among capacitors, reducing thermal stress and improving reliability.

### Capacitance Selection

The required decoupling capacitance is determined based on the worst-case transient current:

$$C = \frac{I_{\text{peak}} \cdot t_{\text{rise}}}{\Delta V} \tag{6.9}$$

where:

- $I_{\text{peak}}$: Maximum transient current

- $t_{\text{rise}}$: MOSFET rise time
- $\Delta V$: Allowable voltage ripple (1–3% of DC bus voltage)

For the designed system:

- Maximum power = 2000 W
- Minimum DC voltage = 40 V
- $I_{\text{peak}}$ = 50 A
- $t_{\text{rise}}$ = 17 ns
- Allowable ripple = 1.2 V

$$C \approx 0.68\,\mu\text{F}$$

To meet this requirement while minimizing ESR and ESL, six capacitors of 0.1 $\mu$F each are connected in parallel:

$$C_{\text{total}} = 0.6\,\mu\text{F}$$

This configuration enables sufficient transient current support with improved high-frequency performance.

**Capacitor Specifications**

Table 6.3: Specifications of the selected decoupling capacitor

| **Parameter** | **Value** |
|---|---|
| Part Number | CGA5L4X7T2W104M160AA |
| Capacitance | 0.1 $\mu$F (100 nF) |
| Voltage Rating | 450 V DC |
| Tolerance | ±20% |
| Dielectric Type | X7T |
| Temperature Range | −55°C to +125°C |
| Package Size | 1206 (3.2 mm × 1.6 mm) |
| ESR (Typical) | Low |
| Mounting Type | Surface Mount (SMD) |

The chosen MLCC capacitors offer low ESR and ESL, making them highly effective for high-frequency decoupling. Their placement close to MOSFETs ensures minimal loop inductance, thereby reducing voltage spikes and LC ringing. The parallel configuration enhances reliability, distributes current stress, and improves thermal performance.

Furthermore, ceramic capacitors are preferred for high-frequency applications due to their fast transient response, while bulk capacitors handle low-frequency energy requirements. This combination ensures stable DC bus operation across the entire switching spectrum.

### 6.3.4 DC Blocking Capacitor (MKP1848H61012JP2)

A DC blocking capacitor is a fundamental component in electronic circuits used to block direct current (DC) while allowing alternating current (AC) to pass. The MKP1848H61012JP2 is one such high-performance capacitor that is commonly applied in various electronic applications. This section delves into its design parameters, operational principles, and practical applications.

**Technical Specifications**

The MKP1848H61012JP2 capacitor is characterized by its polypropylene film construction, high voltage rating, and low loss factor. Key specifications include:

- Capacitance: $10\,\mu$F
- Voltage Rating: 1200 VDC
- Tolerance: ±5%
- Temperature Range: $-40\,^{\circ}$C to $+85\,^{\circ}$C
- Dielectric Material: Metallized Polypropylene (MKP)

These specifications make it suitable for high-frequency and high-voltage applications. Capacitors exhibit a reactance given by:

$$X_C = \frac{1}{2\pi f C}$$

where $X_C$ is the capacitive reactance (in Ω), $f$ is the frequency of the signal (Hz), and $C$ is the capacitance (in F). At DC ($f = 0\,$Hz), $X_C$ approaches infinity, blocking current. For higher frequencies, the reactance decreases, allowing AC signals to pass.

### 6.3.5 Inductor (74435581200)

One key parameter in DAB converter design is leakage inductance, which influences power transfer efficiency, soft-switching operation, and overall system performance. This section analyzes the leakage inductance characteristics of the 74435581200 in DAB converters and enables design considerations for optimal performance.

Table 6.4: Technical Specifications of 74435581200 Inductor

| Parameter | Value |
|---|---|
| Part Number | 74435581200 |
| Inductance | 12 $\mu$H |
| DC Resistance (DCR) | 4.3 m$\Omega$ max |
| Saturation Current | 25 A |
| Rated Current | 19 A |
| Core Material | Ferrite |
| Shielding | Magnetic Shielding |
| Leakage Inductance | Typically Low |
| Operating Temperature | $-40\,^{\circ}$C to $+125\,^{\circ}$C |
| Package Type | Surface Mount (SMD) |

**Specifications**

- **Inductance Value:** Determines the energy storage capability and filtering efficiency.
- **Saturation Current:** Should be greater than the peak current in the circuit.
- **DC Resistance (DCR):** Lower values reduce power losses.
- **Thermal Considerations:** Must operate within temperature limits to ensure reliability.
- **Physical Size and Mounting:** The SMD package makes it suitable for compact designs.

## 6.4 Gate Driver Circuit

### 6.4.1 Gate Driver (UCC21520DW)

Gate driver ICs are essential components in power electronics circuits that drive MOSFETs and IGBTs in applications such as DC-DC converters, motor drives, and industrial power systems. The UCC21520DW is a dual-channel, isolated gate driver developed by Texas Instruments, offering high efficiency, fast switching, and reinforced isolation.

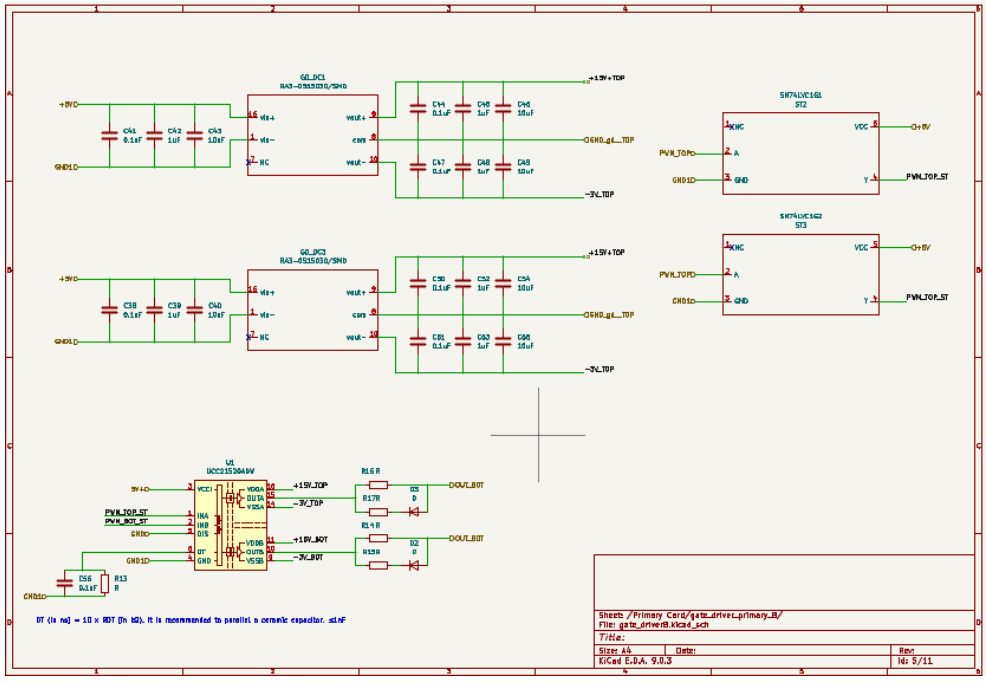


Figure 6.1: Gate driver circuit schematic

### Technical Specifications

The UCC21520DW is a high-speed, dual-channel gate driver that isolates and amplifies the control signal from a microcontroller (MCU) or PWM controller to drive power transistors (MOSFETs/IGBTs) efficiently.

**Signal Isolation and Safety:** The IC uses reinforced isolation technology, allowing it to provide galvanic isolation between the input control signals and the output gate drive signals. This ensures:

- Protection against high-voltage transients.
- Reduced electromagnetic interference (EMI).
- Safer operation in high-power circuits.

**Dead Time Control:** The UCC21520DW features programmable dead-time control, preventing cross-conduction in half-bridge topologies. The dead-time ($t_d$) can be calculated as:

$$t_d = R_d \times C_d$$

where $R_d$ is the dead-time programming resistor (Ω) and $C_d$ is the dead-time capacitor.

Table 6.5: Technical Specifications of UCC21520DW

| **Parameter** | **Value** |
|---|---|
| Part Number | UCC21520DW |
| Channels | 2 (Dual) |
| Isolation Voltage | 5.7 kV RMS |
| Propagation Delay | 45 ns |
| Output Peak Current | 4 A (Source), 6 A (Sink) |
| Gate Drive Voltage | −5 V to 15 V |
| Common Mode Transient Immunity (CMTI) | 125 V/ns |
| Dead Time Control | Programmable |
| Package Type | SOIC-16 |

**Design Considerations**

**Selecting the Right Gate Resistor:** The gate resistor ($R_g$) determines the switching speed and power dissipation of the MOSFET or IGBT. It can be calculated using:

$$R_g = \frac{V_{gate}}{I_{peak}}$$

where $V_{gate}$ is the gate drive voltage (V) and $I_{peak}$ is the peak gate drive current (A).

**Thermal Considerations:** The UCC21520DW must operate within its thermal limits to ensure reliability. The power dissipation can be estimated as:

$$P = V_{DD} \times I_{DD}$$

where $P$ is the power dissipation (W), $V_{DD}$ is the supply voltage (V), and $I_{DD}$ is the supply current (A).

### 6.4.2 Buffer IC (SN74LVC1G17QDBVRQ1)

The SN74LVC1G17QDBVRQ1 implements a Schmitt trigger input, which introduces hysteresis into the input signal transition. This helps reject slow or noisy signals that might

otherwise cause false switching:

$$V_{IH} > V_{IL} \implies \text{Hysteresis} = V_{IH} - V_{IL}$$

This voltage hysteresis ensures clean rising and falling edges, even if the input is noisy or slowly changing. As a buffer, it enables:

- Drive capability to source/sink current into capacitive loads (e.g., gate drivers).
- Signal isolation between logic control and driver stages.
- Short propagation delay (8 ns), suitable for high-speed switching.

Table 6.6: Specifications of SN74LVC1G17QDBVRQ1

| **Parameter** | **Value** |
|---|---|
| Part Number | SN74LVC1G17QDBVRQ1 |
| Function | Single Buffer (Schmitt Trigger) |
| Voltage Range | 1.65 V to 5.5 V |
| Input Type | CMOS / TTL Compatible |
| Output Drive | ±32 mA at 4.5 V |
| Propagation Delay | 8 ns (typ. at 3.3 V) |
| Hysteresis (typ.) | 800 mV |
| Operating Temp Range | −40 °C to +125 °C |
| Package Type | SOT-23-5 (DBV) |
| Automotive Qualified | Yes (AEC-Q100) |

**Application in Gate Driver Circuits**

**Purpose in Gate Driver Front-End:** In a typical setup, the SN74LVC1G17QDBVRQ1 is placed between a microcontroller's PWM output and the input pin of an isolated gate driver such as the UCC21520DW.

**Benefits:**

- Improves signal integrity before it enters the isolation barrier.

- Prevents false triggering due to signal bounce or ringing.
- Buffers MCU outputs that cannot source enough current for fast switching.

### 6.4.3 DC-DC Converter MGJ2D151505BSC for Gate Driver

**Technical Specifications**

Table 6.7: Specifications of MGJ2D151505BSC

| **Parameter** | **Value** |
|---|---|
| Part Number | MGJ2D151505BSC |
| Input Voltage | 15 V DC |
| Output Voltages | +15 V and −5 V (Dual) |
| Output Power | 2 W |
| Isolation Voltage | 5.4 kV RMS |
| Efficiency | 85% (typical) |
| Operating Temperature | −40 °C to +100 °C |
| Package Type | Surface Mount (SMD) |
| Application | Gate Driver Power Supply |

**Functional Description:** The MGJ2D151505BSC is a dual-output isolated DC-DC converter. It enables:

- +15 V output for turning ON the gate of a MOSFET/IGBT.
- −5 V output to ensure fast and safe turn-OFF by keeping the gate well below the source potential.
- Galvanic isolation to protect the low-voltage control side from high-voltage switching noise and transients.

**Role in UCC21520DW Gate Driver Circuit:** The UCC21520DW is a reinforced isolated gate driver used to switch power MOSFETs and IGBTs in high-side/low-side configurations. Each driver channel must be powered by an isolated floating supply, especially for the high-side switch, to:

- Maintain proper gate drive voltage.
- Prevent ground loops and electrical damage.
- Enable bidirectional and high-voltage switching safely.

**MGJ2D151505BSC Integration**

- The +15 V output connects to the VDD of the UCC21520DW driver.
- The −5 V output connects to the VEE pin of the driver.
- The input (15 V) is sourced from a control power rail.
- Proper decoupling capacitors (e.g., 0.1 $\mu$F + 10 $\mu$F) are placed close to VDD and VEE pins.

**Advantages of Using MGJ2D151505BSC:**

- Compact size and SMD footprint for high-density PCB designs.
- High isolation voltage (5.4 kV RMS) ensures safety in high-voltage applications.
- Dual output voltages eliminate the need for discrete bias circuitry.
- Optimized for gate drivers — reduces BOM and simplifies layout.
- Improved noise immunity through isolated construction and proper ground referencing.

### 6.4.4 Turn ON and Turn OFF Resistor (L-0805M5R00FGTS) for Gate Driver Circuit

Gate resistors are essential in controlling the switching characteristics of power semiconductors like MOSFETs and IGBTs. The L-0805M5R00FGTS is a 5 Ω, 0805 surface-mount resistor used as a gate resistor in power driver circuits utilizing the UCC21520DW.

**Purpose of Gate Resistors:** Gate resistors limit the charging and discharging rate of the gate capacitance ($C_{gs}$, $C_{gd}$) in MOSFETs or IGBTs. They:

- Control the gate drive current.
- Adjust switching speed to balance efficiency and EMI.
- Reduce voltage overshoot and ringing.
- Prevent false turn-on due to Miller effect.

**Gate Charge Equation:** The gate charging time is determined by:

$$t_{rise/fall} = R_g \cdot C_{iss}$$

where $R_g$ is the gate resistance (Ω) and $C_{iss}$ is the input capacitance of the MOSFET (F).

Table 6.8: Specifications of L-0805M5R00FGTS

| **Parameter** | **Value** |
|---|---|
| Part Number | L-0805M5R00FGTS |
| Resistance | 5 Ω ±1% |
| Power Rating | 0.2 W |
| Package | 0805 (Surface Mount) |
| Tolerance | ±1% |
| Temperature Coefficient | ±100 ppm/°C |
| Operating Temperature | −55 °C to +155 °C |
| Technology | Thin Film |

- Used between the gate driver output and the MOSFET/IGBT gate terminal.
- Can be split into separate resistors for turn-on and turn-off using diode control paths.
- Turn-on resistor: Connected in series with a diode in forward direction.
- Turn-off resistor: Connected in parallel via reverse-biased diode.
- This enables different speeds for turn-on and turn-off, improving switching performance and reducing EMI.

**Gate Power Loss:**

$$P_{gate} = Q_g \cdot V_{gs} \cdot f_{sw}$$

where $Q_g$ is the gate charge (C), $V_{gs}$ is the gate voltage (V), and $f_{sw}$ is the switching frequency (Hz).

**Resistor Selection:**

- The 5 Ω value balances switching speed and EMI.
- Too low → Fast switching but high EMI and overshoot.
- Too high → Slower switching, more losses, delayed transitions.

### 6.4.5 Diode (PMEG4030AEXEX) for Gate Driver Circuit

The PMEG4030AEXEX is a 40 V, 3 A Schottky barrier diode that plays a crucial role in gate driver circuits by providing fast recovery paths, clamping, and protection functionalities.

Table 6.9: Specifications of PMEG4030AEXEX Diode

| **Parameter** | **Value** |
|---|---|
| Part Number | PMEG4030AEXEX |
| Type | Schottky Barrier Diode |
| Reverse Voltage ($V_R$) | 40 V |
| Forward Current ($I_F$) | 4.2 A |
| Forward Voltage Drop ($V_F$) | 0.45 V (typ at 3 A) |
| Reverse Leakage Current | 25 $\mu$A max at 40 V |
| Recovery Time | Very fast (Schottky type) |
| Package | SOD882 (ultra-small) |

**Split Gate Resistor Network:**

- UCC21520DW driver output → PMEG4030AEXEX anode.
- Cathode connected to gate via turn-on resistor (e.g., 5 Ω).
- In parallel: gate drive return path through a 10 Ω resistor for turn-off.

Although PMEG4030AEXEX is not specifically a bootstrap diode, its low $V_F$ and high-speed switching make it well suited for this role.

## 6.5 Current Sensor Circuit

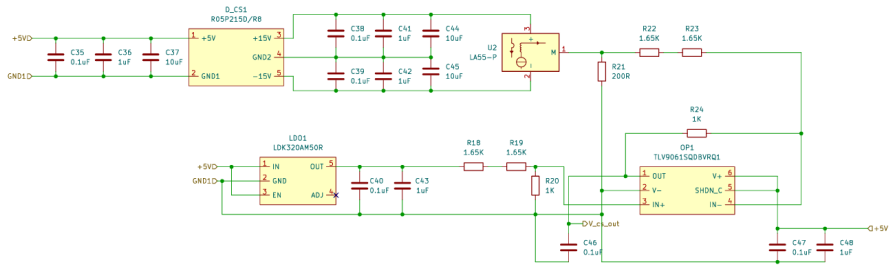



Figure 6.2: Current sensing circuit

### 6.5.1 Current Sensor (LA 55-P)

Accurate current measurement is vital in modern power electronics, motor control, and energy management systems. The LA 55-P from LEM is a high-performance closed-loop Hall-effect current sensor designed for measuring AC, DC, and pulsed currents with high precision and electrical isolation.

**Features:**

- Measures DC, AC, and complex waveforms.
- Galvanically isolated between primary and secondary circuits.
- High bandwidth and fast response.
- Linear output and low offset drift.
- Compact and through-hole package.

**Working Principle:** The LA 55-P operates using the Hall-effect principle in a closed-loop (compensated) architecture. A Hall-effect element placed in a magnetic core senses the magnetic field created by the primary current and generates a voltage proportional to the magnetic field strength. In closed-loop sensors:

- A secondary compensation current is driven in a coil to nullify the magnetic field sensed by the Hall element.

- The compensation current is proportional to the primary current.
- This compensation current is output through a current output pin, making it easy to sense via a burden resistor.

$$I_{out} = \frac{I_{primary}}{1000}$$

This results in high linearity, fast response, and low offset drift.

Table 6.10: Electrical Specifications of LA 55-P

| **Parameter** | **Value** |
|---|---|
| Part Number | LA 55-P |
| Technology | Closed-loop Hall-effect (compensated) |
| Nominal Current ($I_{pn}$) | 50 A |
| Measurement Range | ±50 A |
| Conversion Ratio | 1:1000 (50 A → 50 mA) |
| Supply Voltage | ±15 V (dual supply) |
| Response Time | 1 $\mu$s |
| Accuracy | ±0.9% of full scale |
| Output Type | Analog current (via burden resistor) |
| Operating Temperature | −40 °C to +85 °C |

**Burden Resistor:** The output current is typically passed through an external burden resistor $R_b$ to convert it into a voltage:

$$V_{out} = I_{out} \cdot R_b$$

For example, for a 50 A primary current → 50 mA output, with a 100 Ω resistor → $V_{out}$ = 5 V. A low-pass RC filter can be added to the output to remove high-frequency noise and prevent aliasing in ADCs.

### 6.5.2 Offset LDO (LDK320AM50R)

In modern current sensing circuits, particularly those using a single-supply rail, it is necessary to bias the op-amp input and output signals to a known offset voltage to accommodate

bidirectional current measurement. The LDK320AM50R is a low dropout (LDO) regulator from STMicroelectronics that provides a clean and stable 5.0 V supply used to derive a reference offset voltage.

In single-supply systems (e.g., 0–5 V), the op-amp cannot output negative voltages. To measure bidirectional current, the sensed voltage across a shunt resistor must be biased above ground, often to a mid-supply reference of 2.5 V.

Table 6.11: Key Specifications of LDK320AM50R

| **Parameter** | **Value** |
|---|---|
| Output Voltage | 5.0 V fixed |
| Output Current | 200 mA max |
| Dropout Voltage | 200 mV (typ @ 100 mA) |
| Output Noise | 150 $\mu$V RMS |
| PSRR | 48 dB @ 1 kHz |
| Quiescent Current | 65 $\mu$A (typ) |
| Input Voltage Range | 2.5 V to 18 V |
| Package | SOT-23-5 |

A simple voltage divider from the LDO output creates this reference:

$$V_{REF} = V_{LDO} \times \frac{R_2}{R_1 + R_2}$$

To create a 2.5 V offset:

$$R_1 = R_2 \implies V_{REF} = \frac{5\,\text{V}}{2} = 2.5\,\text{V}$$

This offset is then connected to the non-inverting input of the op-amp and the reference node in differential amplifier designs.

### 6.5.3 Operational Amplifier in Current Sensing Circuits (TLV9061SQDBVRQ1)

Current sensing is a fundamental requirement in modern power electronic systems including DC-DC converters, battery management systems, motor drives, and inverters. Accurate current measurement is critical for system control, protection, and diagnostics. An op-amp

used for current sensing must:

- Accurately amplify small signals (typically millivolts).
- Have low input offset voltage.
- Offer high Common-Mode Rejection Ratio (CMRR).
- Operate with low power and wide supply range.

## 6.6 Voltage Sensing Circuit

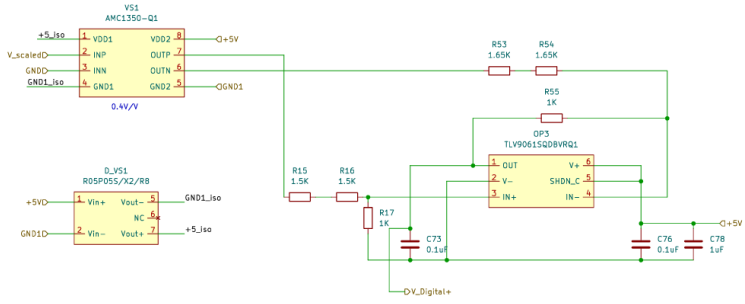



Figure 6.3: Voltage sensing circuit

### 6.6.1 Optically Isolated Amplifier (ACNT-H87A-000E)

In high-voltage systems, it is crucial to sense voltages accurately while maintaining electrical isolation between the sensing and control sides. The ACNT-H87A-000E is a high-speed optocoupler from Broadcom that provides a reliable and compact solution for isolated voltage sensing. This device is particularly suitable for SMPS, motor drives, and industrial automation, where galvanic isolation and signal fidelity are essential.

**Galvanic Isolation:** Voltage sensing circuits often need to interface with high-voltage primary systems and low-voltage microcontrollers or ADCs. Direct connection can damage sensitive control electronics, create ground loops and EMI issues, and violate safety standards.

Table 6.12: Key Electrical Specifications of ACNT-H87A-000E

| Parameter | Value |
|---|---|
| Isolation Voltage | 7500 Vrms |
| Output Type | Open collector |
| Operating Temperature | −40 °C to +110 °C |
| Package Type | Wide-body DIP-8 |

**Role of Optocouplers:** Optocouplers provide electrical isolation, fast noise-immune transmission, and long-term signal integrity across isolation barriers.

**Principle of Operation:** The ACNT-H87A-000E consists of an LED that emits infrared light when forward biased and a photodetector (phototransistor) that turns ON in response to light, forming a photoelectric interface that transmits a digital signal while keeping input and output galvanically isolated.

**Working Description:**

- A comparator converts analog voltage to a PWM signal.
- The PWM signal drives the input of ACNT-H87A-000E.
- The optocoupler output is pulled up and read by a microcontroller/ADC.
- Duty cycle or frequency is decoded as proportional to the sensed voltage.

### 6.6.2 DC-DC Converter (R1SE-3.305/H2-R) for Voltage Sensor

To power the isolated side of the voltage sensor, a DC-DC converter with isolation capability is required. This ensures that the primary and secondary grounds remain electrically separated, preserving signal integrity and operator safety.

Table 6.13: Typical Specifications of a 5 V Isolated DC-DC Converter

| Parameter | Value |
|---|---|
| Input Voltage | 3.3 V |
| Output Voltage | 5 V isolated |
| Output Current | 200 mA max |
| Isolation Voltage | 2 kV |
| Package | SIP or SMD |

### 6.6.3 OPAMP (OPA237UA) for Voltage Sensor Circuit

The OPA237UA from Texas Instruments is a dual high-precision op-amp known for its low offset voltage, low bias current, and excellent common-mode rejection.

Table 6.14: OPA237UA Key Electrical Characteristics

| Parameter | Value |
|---|---|
| Input Bias Current | 40 nA |
| Supply Voltage Range | ±1.35 V to ±18 V |
| Gain Bandwidth Product | 1.5 MHz |
| Slew Rate | 0.3 V/$\mu$s |
| Common-Mode Rejection Ratio (CMRR) | 500 mV/µs |
| Power Supply Rejection Ratio (PSRR) | 100 dB |

**Amplifier Configurations:**

- Buffer (Unity Gain): $V_{out} = V_{in}$
- Non-Inverting Amplifier:
$$V_{out} = V_{in} \cdot \left(1 + \frac{R_2}{R_1}\right)$$
- Differential Amplifier:
$$V_{out} = \frac{R_2}{R_1} \cdot (V_2 - V_1)$$

**Why OPA237UA:**

- Ultra-low offset ensures accuracy over temperature.
- High CMRR minimizes noise from power supply or common ground.
- A resistor divider scales down high voltage; the OPA237UA buffers and amplifies this signal before passing it to an ADC or comparator.

### 6.6.4 Loss Calculation

#### Conduction Loss

The conduction loss $P_{cond}$ in the MOSFETs can be estimated using:

$$P_{cond} = I_{rms}^2 \times R_{DS(on)} \tag{6.10}$$

Assuming balanced current sharing and based on VTDAB operation:

$$I_{rms} \approx \frac{P_{out}}{V_{primary}} = \frac{1000\,\text{W}}{100\,\text{V}} = 10\,\text{A}$$

$$P_{cond} = (10\,\text{A})^2 \times 25 \times 10^{-3}\,\Omega = 2.5\,\text{W} \quad \text{per MOSFET}$$

#### Switching Loss

The switching loss $P_{sw}$ per device can be estimated as:

$$P_{sw} = \frac{1}{2} V_{DS}\, I_{load}\, (t_{on} + t_{off})\, f_{sw} \tag{6.11}$$

where $V_{DS} = 100\,\text{V}$, $I_{load} = 10\,\text{A}$, $t_{on} + t_{off} \approx 100\,\text{ns}$, and $f_{sw} = 50\,\text{kHz}$. Thus:

$$P_{sw} = \frac{1}{2} \times 100 \times 10 \times 100 \times 10^{-9} \times 50 \times 10^3 = 2.5\,\text{W} \quad \text{per MOSFET}$$

$$P_{total} = P_{cond} + P_{sw} = 2.5\,\text{W} + 2.5\,\text{W} = 5\,\text{W}$$

# Chapter 7

# Transformer design of VTDAB Converter

## 7.1 Introduction

The Variable-Turns Dual Active Bridge (VTDAB) converter represents a significant advancement in isolated DC–DC power conversion for wide-voltage-range applications. Unlike the conventional single-transformer Dual Active Bridge (DAB), the VTDAB topology employs two transformers with distinct turns ratios, enabling the converter to maintain efficient operation across a wide secondary voltage range—in this case, 40 V to 250 V—from a fixed 100 V primary input at a rated power of 1 kW and switching frequency of 50 kHz.

The transformer is the most critical magnetic component in any DAB-derived converter. Its design directly governs conduction losses (copper loss), magnetic losses (core loss), leakage inductance (which serves as the energy-transfer element in DAB operation), ZVS range, and overall power density. A poorly designed transformer can cause excessive core saturation, high RMS currents, poor thermal performance, and failure to achieve zero-voltage switching (ZVS).

In the VTDAB, the two transformers are switched into the circuit depending on the output voltage range required, allowing the converter to maintain a near-unity effective turns ratio at any operating point. This chapter presents the complete magnetic design procedure for both transformers—from core selection and turns calculation to winding design and the final fabricated specifications received from the manufacturer.

## 7.2 Electrical Specifications

The transformer design is governed by the electrical specifications of the VTDAB converter. These specifications define the input/output voltage range, power level, and switching frequency that the magnetic components must support. Table 7.1 summarises the key electrical parameters.

Table 7.1: Electrical specifications of the VTDAB converter

| **Parameter** | **Symbol** | **Value** | **Unit** |
|---|---|---|---|
| Primary (Input) Voltage | $V_1$ | 100 | V |
| Secondary (Output) Voltage Range | $V_2$ | 40–250 | V |
| Rated Power | $P_{\text{rated}}$ | 1000 | W |
| Maximum Power | $P_{\text{max}}$ | 2000 | W |
| Nominal Input Current | $I_1$ | 10 | A |
| Output Current Range | $I_2$ | 4–25 | A |
| Switching Frequency | $f_s$ | 50 | kHz |
| Converter Topology | – | VT-DAB | – |

The wide secondary voltage range (40 V to 250 V, a 6.25 : 1 ratio) poses a fundamental challenge for a single fixed-ratio transformer: the effective turns ratio cannot be simultaneously optimal for both extreme operating points. The VTDAB architecture resolves this by using two transformers—Transformer 1 (T1) for the low-voltage range (40–73 V) and Transformer 2 (T2) for the high-voltage range (123–250 V). This strategy improves efficiency across the entire operating envelope.

## 7.3 Transformer Design Theory and Constraints

The design of high-frequency transformers requires balancing two competing loss mechanisms: core loss ($P_{fe}$) and copper loss ($P_{cu}$). The total power dissipation is:

$$P_{\text{tot}} = P_{fe} + P_{cu} \tag{7.1}$$

Core loss increases with the peak AC flux density swing $\Delta B$, while copper loss decreases with increasing $\Delta B$ (since fewer turns are needed). This trade-off defines an optimal

operating flux density that minimizes total loss.

### 7.3.1 Core Loss Model

Core loss is modelled using the Steinmetz-based expression:

$$P_{fe} = K_{fe} \cdot (\Delta B)^{\beta} \cdot A_c \cdot l_m \tag{7.2}$$

where $K_{fe}$ is a frequency-dependent core loss coefficient, $\beta$ is the Steinmetz exponent, $A_c$ is the effective cross-sectional area of the core, and $l_m$ is the mean magnetic path length. The product $A_c \cdot l_m$ represents the effective core volume. For the selected core material (Material H, $f > 20\,\text{kHz}$), the parameters are:

- $k = 5.372 \times 10^{-5}$
- $m = 1.62$ (frequency exponent)
- $n = \beta = 2.15$ (flux density exponent)

The effective core loss coefficient is obtained by scaling the manufacturer's specific loss figure (W/kg) by core density:

$$K_{fe} = k \cdot f^{m} \cdot \rho_{\text{core}} \tag{7.3}$$

$$K_{fe} = 5.372 \times 10^{-5} \times (5 \times 10^{4})^{1.62} \times 5.24 \times 10^{-3} \approx 11.53 \ [\text{W/cm}^3 \cdot \text{T}^{\beta}] \tag{7.4}$$

### 7.3.2 Copper Loss Model

The copper loss is minimised when the window area $W_A$ is optimally allocated among primary and secondary windings. The expression for copper loss, as a function of $\Delta B$, is:

$$P_{cu} = \left[ \rho \cdot \lambda_1^2 \cdot I_{\text{tot}}^2 \cdot \frac{4K_u}{(W_A \cdot A_c^2)} \cdot \frac{\text{MLT}}{(\Delta B)^2} \right] \tag{7.5}$$

where $\rho$ is the resistivity of conductor (copper, $1.724 \times 10^{-6}\,\Omega \cdot \text{cm}$), $\lambda_1$ is the applied volt-seconds on the primary, $I_{\text{tot}}$ is the total RMS current referred to the primary, $K_u$ is the window utilisation factor, and MLT is the mean length per turn.

### 7.3.3 Optimum Flux Density

Setting $\frac{dP_{\text{tot}}}{d(\Delta B)} = 0$ yields the optimum flux density:

$$\Delta B_{\text{opt}} = \left[ \frac{\rho \cdot \lambda_1^2 \cdot I_{\text{tot}}^2 \cdot (2K_u)}{(\beta \cdot K_{fe}) \cdot \left( \frac{\text{MLT}}{W_A \cdot A_c^3 \cdot l_m} \right)} \right]^{\frac{1}{\beta+2}} \tag{7.6}$$

### 7.3.4 Core Geometrical Constant $K_{gfe}$

The core selection criterion is expressed through the core geometrical constant $K_{gfe}$, which encapsulates the dimensional properties of the core:

$$K_{gfe} = W_A \cdot A_c^{2(\beta-1)/\beta} \cdot \text{MLT}^{2/\beta} \cdot l_m^{2/\beta} \left[ \left( \frac{\beta}{2} \right)^{-\beta/(\beta+2)} + \left( \frac{\beta}{2} \right)^{2/(\beta+2)} \right]^{-(\beta+2)/\beta} \tag{7.7}$$

For a valid design, the chosen core must satisfy:

$$K_{gfe} \geq \frac{\rho \cdot \lambda_1^2 \cdot I_{\text{tot}}^2 \cdot K_{fe}^{2/\beta}}{4K_u \cdot P_{\text{tot}}^{(\beta+2)/\beta}} \times 10^8 \tag{7.8}$$

## 7.4 Flux Density Selection for the VTDAB Topology

Flux density selection in VTDAB operation differs from standard transformer topologies due to several unique operating conditions imposed by the dual-active-bridge modulation scheme:

- **Square-wave excitation:** Unlike sinusoidal transformers, the DAB applies square

voltage waveforms, resulting in a trapezoidal flux waveform with steep transitions.

- **Bidirectional power flow:** The transformer must support flux swing in both directions every half cycle, resulting in full $\Delta B$ swing (from $-B_{\text{peak}}$ to $+B_{\text{peak}}$) each period.
- **High RMS flux swing:** The bidirectional flux operation increases hysteresis loss compared to unidirectional topologies.
- **DC bias during transient phase shifts:** During load transient or operating mode transitions in VTDAB, temporary imbalances in the volt-second product can cause DC flux offset, partially biasing the core toward saturation.
- **Multiple operating modes:** The VTDAB may operate in single-phase-shift, extended-phase-shift, or dual-phase-shift modes depending on the voltage conversion ratio, introducing asymmetry in the applied volt-seconds.

Due to these factors, the VTDAB design conservatively operates at a lower flux density than is typically chosen for single-direction topologies. While conventional power transformers in other topologies commonly operate at $B_{\text{peak}} = 0.2\,\text{T}$, and the ferrite material used (N87) has a saturation flux density of approximately 490 mT at 25°C and 390 mT at 100°C, the VTDAB transformers are designed to operate at:

$$B_{\text{operating}} = 0.128\,\text{T (T1)}, \quad 0.1256\,\text{T (T2)} \tag{7.9}$$

This conservative choice—approximately 0.13–0.15 T provides margin against asymmetric flux drift, reduces core loss at 50 kHz, and ensures reliable ZVS operation across the full operating envelope.

## 7.5 Core Material and Shape Selection

### 7.5.1 Ferrite Core Material — N87 (TDK/EPCOS)

The N87 grade MnZn power ferrite material from TDK/EPCOS was selected as the core material for both VTDAB transformers. N87 is specifically optimised for power conversion applications in the 25 kHz to 200 kHz range. Its key magnetic properties are listed in Table 7.2.

Table 7.2: Key properties of N87 MnZn ferrite material

| **Property** | **Value** | **Condition** |
|---|---|---|
| Saturation Flux Density ($B_s$) | 490 mT | 25°C |
| Saturation Flux Density ($B_s$) | 390 mT | 100°C |
| Initial Permeability ($\mu_i$) | $\sim$ 2200 | 25°C |
| Minimum Core Loss | $\sim$ 25 kW/m$^3$ (typ.) | 100 kHz, 200 mT, 100°C |
| Curie Temperature | $>$ 210°C | – |
| Optimal Frequency Range | 25–200 kHz | Power applications |

The selection of N87 is justified by its exceptionally low core loss in the 50 kHz operating range, stable permeability over the expected operating temperature range (20–100°C), and wide linear B–H region which accommodates the bidirectional flux swing of the VTDAB without entering saturation. The 3C95 material from Ferroxcube was also considered ($B_s$ = 530 mT at 25°C, very flat loss-temperature curve), but N87 was preferred due to availability and established design data.

### 7.5.2 Core Shape — EE Core (Shell Type)

An EE-type (shell-type) ferrite core was selected. In the EE configuration, the windings are placed on the centre limb, and the two outer limbs serve as the magnetic flux return paths. This arrangement results in an enclosed magnetic path with a small window area, which is well-suited to concentric winding configurations. Compared to core-type geometries (CC/CI), the EE core offers:

- Lower leakage flux due to the enclosed winding window.
- Good EMI containment as the flux is largely confined within the core.
- Mechanical simplicity — two identical E-halves are assembled around the bobbin.
- Wide availability of standardised sizes and compatible bobbins.

The winding configuration adopted is concentric non-interleaved for T2 (low leakage target $< 2.1\,\mu$H) and concentric for T1 (leakage $< 10.2\,\mu$H acceptable as it aids phase-shift energy transfer). In concentric winding, the primary and secondary are placed on the same centre limb of the EE core, separated by insulating tape.

### 7.5.3 Core Size — EE6527 (ETD44-class)

The core size was determined through the $K_{gfe}$-based design procedure. With the following input parameters:

- $\rho$ (copper resistivity) = $1.724 \times 10^{-6}\,\Omega \cdot \text{cm}$
- $\lambda_1$ (primary volt-seconds) = $\frac{V_1}{2f_s} = \frac{100}{2\times50\times10^3} = 10^{-3}\,\text{V} \cdot \text{s}$
- $I_{\text{tot}}$ (total RMS current referred to primary) = 40 A (at 2 kW)
- $K_u$ (window utilisation factor) = 0.3
- $P_{\text{tot}}$ (allowed transformer loss) = 10 W
- $\beta = 2.15, \quad K_{fe} \approx 11.53\,\text{W/cm}^3 \cdot \text{T}^{\beta}$

The required core geometrical constant evaluates to:

$$K_{gfe} \geq 2.624 \times 10^{-3}\,\text{cm}^5 \text{ (approx.)} \tag{7.10}$$

Consulting the ETD ferrite core table (Table 7.3), the ETD44 core satisfies this requirement and was selected as the design basis core. The final fabricated transformers were wound on the EE6527 core, which is closely equivalent in geometry and window area to the ETD44 series.

Table 7.3: ETD ferrite core geometric parameters. ETD44 selected (✓) as it meets $K_{gfe}$ requirement.

| **Part No.** | **MLT (cm)** | $l_m$ **(cm)** | $W_A$ **(cm$^2$)** | $A_c$ **(cm$^2$)** | $W_A A_c^2$ **(cm$^5$)** | $K_{gfe}$ **(cm$^5$)** |
|---|---|---|---|---|---|---|
| ETD-29 | 6.4 | 7.20 | 1.865 | 0.761 | 1.419 | 0.0517 |
| ETD-34 | 7.1 | 7.87 | 1.757 | 0.974 | 1.711 | 0.0911 |
| ETD-39 | 8.3 | 9.22 | 1.757 | 1.252 | 2.200 | 0.1766 |
| ETD-44 (✓) | 9.4 | 10.30 | 1.599 | 1.742 | 2.785 | 0.3595 |
| ETD-49 | 10.3 | 11.40 | 1.627 | 2.110 | 3.434 | 0.5917 |
| ETD-54 | 11.7 | 12.70 | 1.609 | 2.800 | 4.505 | 1.2104 |

## 7.6 Turns Calculation — Theoretical Design

### 7.6.1 Primary Turns from EMF Equation (E55 Core Reference)

An initial design was also carried out using the E55/28/21 (N87) ferrite core with $A_c = 250\,\text{mm}^2 = 250 \times 10^{-6}\,\text{m}^2$, $B_{\max} = 0.2\,\text{T}$. Applying the transformer EMF equation:

$$N_p = \frac{V_p}{4.44 \cdot f_s \cdot B_{\max} \cdot A_c} \tag{7.11}$$

$$N_p = \frac{100}{4.44 \times 50 \times 10^3 \times 0.2 \times 250 \times 10^{-6}} \approx 9 \text{ turns} \tag{7.12}$$

This enables the minimum primary turns to avoid core saturation at $B_{\max} = 0.2\,\text{T}$. The 4.44 factor corresponds to the RMS form factor of a square wave ($4/\sqrt{(8/\pi^2)} \approx 4.44$).

### 7.6.2 Primary Turns from Optimum Design Procedure (ETD44 Core)

Using the optimum flux density $\Delta B_{\text{opt}} = 0.2237\,\text{T}$ derived from the total loss minimisation procedure with the ETD44 core ($A_c = 1.742\,\text{cm}^2$):

$$n_1 = \frac{\lambda_1}{2 \cdot \Delta B \cdot A_c} \times 10^4 \tag{7.13}$$

$$n_1 = \frac{10^{-3}}{2 \times 0.2237 \times 1.742} \times 10^4 \approx 12.8 \rightarrow 12 \text{ turns} \tag{7.14}$$

This value is consistent with the fabricated transformer specifications which use 11–12 turns on the primary, validating the design methodology.

### 7.6.3 Turns Ratios and Secondary Turns

Two distinct turns ratios are required to cover the full output voltage range of 40–250 V:

Table 7.4: Design summary of turns ratios for VTDAB transformers

| Transformer | Output Voltage Range | Turns Ratio $n = N_s/N_p$ | Primary Turns $N_p$ | Secondary Turns $N_s$ |
|---|---|---|---|---|
| T1 (Low-side) | 40–73 V | $n_1 \approx 0.556$ (6/11) | 9–11 | 5–6 |
| T2 (High-side) | 123–250 V | $n_2 \approx 1.5$ (18/12) | 9–12 | 13–18 |

## 7.7 Winding Design

### 7.7.1 Wire Area Calculation

The window area of the core must be shared between primary and secondary windings. The fraction allocated to each winding is:

$$\alpha_j = \frac{n_j \cdot I_j}{n_1 \cdot I_{\text{tot}}} \tag{7.15}$$

For equal power distribution between primary and secondary sides: $\alpha_1 = \alpha_2 = 0.5$. The required wire cross-sectional area for each winding is:

$$A_{w1} \leq \frac{0.5 \times 0.3 \times 1.599}{12} \approx 0.020\,\text{cm}^2 \tag{7.16}$$

$$A_{w2} \leq \frac{0.5 \times 0.3 \times 1.599}{18} \approx 0.013\,\text{cm}^2 \tag{7.17}$$

### 7.7.2 Litz Wire Selection

At 50 kHz, solid conductors suffer severe skin effect and proximity effect losses. The skin depth in copper at 50 kHz is:

$$\delta = \sqrt{\frac{\rho}{\pi f \mu_0}} \approx 0.297\,\text{mm} \tag{7.18}$$

To maintain effective conduction, the individual strand diameter must remain well below 0.594 mm.Litz wire comprising multiple individually enamelled thin strands twisted/braided together. Therefore mandatory for high-frequency transformer windings. The use of Litz

wire provides:

- Reduced AC resistance: strands are individually insulated and transposed, equalising current distribution among strands.
- Lower proximity effect losses compared to multi-layer solid conductors.
- Improved thermal performance due to better heat distribution.
- Flexibility for bobbin winding without mechanical stress on individual strands.

For both transformers, 42# Litz wire with 0.1 mm strand diameter is specified. This strand diameter is comfortably below the skin depth limit at 50 kHz. The bundle cross-sectional areas selected are 5 $mm^2$ (primary, 20 A) and 5–11 $mm^2$ (secondary, depending on current).

### 7.7.3 Bobbin and Insulation

The bobbin for the EE6527 core is constructed from flame-retardant polyamide (PA66), offering high dielectric strength (typically 30–35 kV/mm), resistance to thermal deformation up to 180°C, and a UL 94V-0 flammability rating. The bobbin serves multiple functions:

- Provides a structured winding former that ensures uniform turn distribution.
- Maintains physical separation between primary and secondary windings through a central partition, minimising inter-winding capacitance.
- Offers pin terminals for reliable solder or crimp connections to PCB or flying leads.
- Maintains creepage and clearance distances required by IEC 60950 / IEC 62368 safety standards.

Inter-winding tape insulation is applied between the primary and secondary layers. Given the moderate voltage difference in this design ($\leq$ 250 V primary-to-secondary), a single layer of 0.05 mm polyester tape is used as insulation. For higher-voltage isolation requirements, multiple layers of reinforced tape or triple-insulated wire would be required.

## 7.8 Measured Electrical Parameters

Following receipt of the fabricated transformers, electrical measurements were performed to characterise the parasitic parameters critical to DAB converter operation. These include the magnetising inductance ($L_p$), leakage inductance ($Llk$), and DC winding resistance $Rdc$. These parameters are essential inputs to the VTDAB simulation model and experimental validation setup.

Table 7.5: Measured electrical parameters

| Parameter | Value | Unit |
|---|---|---|
| $L_p$ | 1.425 | mH |
| $L_{lk}$ | 3.18 | $\mu H$ |
| $R_p$ | 24.09 | mΩ |
| $R_s$ | 109.13 | mΩ |

The measured leakage inductance of T1 ($Llk = 3.18H$) is well within the specified limit of less than 10.2 µH. In the DAB converter, leakage inductance plays the role of the series inductor through which energy is transferred between the two bridges. The leakage inductance must be carefully managed too high value limits power transfer and ZVS range too low value results in high peak currents. The measured value for T1 is suitable for the phase-shift modulation scheme employed in the VTDAB

The magnetising inductance $L_p = 1.425mH$ is sufficiently large that the magnetising current contributes negligibly to the primary RMS current at 50 kHz, validating the ideal transformer assumption used in the design model.

## 7.9 Comparison: Theoretical Design vs. Fabricated Specifications

Table 7.6 provides a direct comparison between the theoretical design values computed in this chapter and the final fabricated transformer specifications.

Table 7.6: Comparison of theoretical and fabricated transformer parameters

| **Parameter** | **Theoretical (T1)** | **Fabricated (T1)** | **Theoretical (T2)** | **Fabricated (T2)** |
|---|---|---|---|---|
| Primary Turns $N_p$ | 9 (EMF) / 12 (opt.) | 11 | 9 / 12 | 12 |
| Secondary Turns $N_s$ | 5 (EMF) / ~7 (opt.) | 6 | 14 / 18 | 18 |
| Turns Ratio $n$ | 0.556 | 0.545 (6/11) | 1.5 | 1.5 (18/12) |
| Core | E55 / ETD44 | EE6527 | E55 / ETD44 | EE6527 |
| Flux Density (T) | 0.2 (design max) | 0.128 (operating) | 0.2 (design max) | 0.1256 (operating) |
| Wire Type | Litz (calc.) | 42# Litz, 5–11 sq.mm | Litz (calc.) | 42# Litz, 5 sq.mm |
| Frequency | 50 kHz | 50 kHz | 50 kHz | 50 kHz |

## 7.10 Summary

This chapter has presented the complete magnetic design process for the two high-frequency transformers used in the Variable-Turns Dual Active Bridge (VTDAB) converter. The key findings and design outcomes are summarised below:

- The VTDAB converter requires two transformers with distinct turns ratios to efficiently span the 40–250 V output range from a fixed 100 V input at 50 kHz.
- A systematic design procedure based on the trade-off between core loss and copper loss was employed to determine the optimum flux density ($\Delta B_{\text{opt}} \approx 0.224\,\text{T}$ for the ETD44 core) and primary turns ($n_1 \approx 12$).
- The VTDAB topology imposes unique constraints — bidirectional flux, DC bias risk during transients, and asymmetric phase-shift modulation — that necessitate a conservative operating flux density of $\approx 0.13\,\text{T}$, well below the material's saturation limit.
- N87 MnZn ferrite was selected as the core material for its low loss at 50 kHz, stable permeability over temperature, and 0.49 T saturation flux density at 25°C.
- EE-type (shell-type) cores with concentric winding were adopted. The final fabricated core is EE6527, equivalent to the ETD44 design target.
- 42# Litz wire (0.1 mm strand diameter) was specified for all windings to mitigate skin and proximity effects at 50 kHz.

- The fabricated transformers closely match the theoretical design targets, with T1 achieving 11:6 turns ($n \approx 0.545$, $B_{\text{op}} = 0.128\,\text{T}$, $L_{lk} < 10.2\,\mu H$) and T2 achieving 12:18 turns ($n = 1.5$, $B_{\text{op}} = 0.1256\,\text{T}$, $L_{lk} < 2.1\,\mu H$).
- Measured parameters confirm $L_p = 1.425\,\text{mH}$ and $L_{lk} = 3.18\,\mu H$ for T1, suitable for DAB phase-shift energy transfer.

These transformers will be integrated into the VTDAB prototype, and their performance will be validated through experimental measurements of efficiency, thermal rise, and ZVS range in subsequent chapters.

# Chapter 8

# conclusion and future scope

### 8.0.1 Conclusion

: This thesis presents a robust, efficient, and electrically isolated VTDAB converter for second-life battery (SLB) energy storage system applications. Components were carefully selected, and design considerations for magnetics, heat sinks, and signal conditioning were thoroughly addressed. Simulation results validate wide voltage range operation with ZVS maintenance across all operating modes, confirming high efficiency and system safety.

### 8.0.2 Future Scope

: Future work involves integrating the VTDAB with a second-life battery Energy Storage System (SLB-ESS) and connecting it to the grid via a Voltage Source Inverter (VSI), enabling bidirectional energy exchange for grid stabilization, peak shaving, and extended utilization of retired EV batteries.

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